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Geography of Colombian Life Satisfaction

Monday 26th October, 2026 19:41

This is the first study of the geography of Colombian life satisfaction or Subjective Wellbeing (SWB). We use 2023 Encuesta Nacional de Calidad de Vida (ECV), a large (non-missing n>100k) survey by Colombian government. We find quite large differences close to 1 point on 0-10 life satisfaction scale: central parts are the happiest: 'Cafetero' and Santander, and the least happy are outer departments: Western Choco, Southern Narino, and Eastern Guainia and Vichada. We also map life satisfaction across over 40 municipalities and find differences slightly over 1 SWB point. Farmers and farming communities are happier for respondents in 2-6 income deciles, thus confirming 'happy peasant hypothesis,' but in the top decile, non-farmers and non-farming communities are significantly happier–the differences within income deciles are small upto 0.2 SWB point. Finally, we find a strong and linear relationship between safety and life satisfactions. Safety is important in unsafe (and happy) Colombia.

Happiness, Life Satisfaction, Subjective Wellbeing, Colombia, Encuesta Nacional de Calidad de Vida

One of the puzzles in the happiness literature is Latin American phenomenon (Rojas 2015, 2019): high happiness[1] despite low development in Latin America. Colombia is one of the happiest countries in the world–see table 1. But we know little about geography of happiness within Colombia. This paper is about geography of Colombian happiness across its geographies: departments, municipalities, and farmers v non-farmers.

Table 1: 10 happiest countries in the world. Data from World Database of Happiness (WDH) 2010-2019 out of 160 countries at `worlddatabaseofhappiness.eur.nl/rank-reports/satisfaction-with-life`; and World Values Surveys (WVS) 2005-2022 [waves 5-7] out of 88 countries at `https://www.worldvaluessurvey.org`.

WDH			WVS		
rank	country	happiness (1-10)	rank	country	happiness (1-10)
1	Denmark	8.2	1	Puerto Rico	8.4
2	Mexico	8.1	2	Mexico	8.3
3	Colombia	8.1	3	Colombia	8.3
4	Switzerland	8	4	Qatar	8.0
5	Finland	8	5	Norway	7.9
6	Iceland	8	6	Nicaragua	7.9
7	Costa Rica	7.9	7	Tajikistan	7.9
8	Norway	7.9	8	Switzerland	7.9
9	Canada	7.9	9	Uzbekistan	7.9
10	Qatar	7.8	10	Ecuador	7.8

A Peruvian social psychologist specializing in happiness gives a persuasive explanation of (unexpectedly) high Latin happiness:

> the minimalist well-being lessons of Andean and Amazonian small traditional communities which constitute the grounds of Latin American happiness, a life style that mimics the ancestral environment, the deep nature where the happiness brain wiring occurred; a physical and social environment that naturally activates the brain pleasure circuits. Culture resemble evolutionary needs; resources to achieve needs are available for everyone; positive, interdependent collectivist interaction is ingrained in behavior, supporting, working, competing, and sharing. (Yamamoto 2016, p.45)

[1] Following usual practice in social indicators research terms happiness and life satisfaction are used interchangeably, and specifically we mean cognitive and evaluative life satisfaction rather than affective happiness.

There is a Latin American concept of 'buen vivir' (good living) that is about living close to nature and other people, in balance and sustainably, rejecting extractive materialistic and consumerist capitalism (Hidalgo-Capitán and Cubillo-Guevara 2017, Calisto Friant and Langmore 2015, Guardiola and García-Quero 2014, García-Quero and Guardiola 2018). Guardiola and García-Quero (2014), García-Quero and Guardiola (2018) are probably the most relevant here and they are also next door, in Ecuador.

Nature and community contribute to SWB, not just employment income and consumption (Guardiola and García-Quero 2014)–something largely forgotten in the US (Okulicz-Kozaryn and Martinez 2025). For instance, households that grow their own food and are in an indigenous community depend less on money to be happy (García-Quero and Guardiola 2018). Then we would expect that high happiness in Colombia will be found not only in the richest urban parts of the country but also in natural communal rurality. There could be some urban buen vivir especially if city is not too large or dense and with good access to nature, and there could be no buen vivir in rural areas with industrial extractive agriculture and other problems. But in general probably the more rural, the more nature, community, and buen vivir.

1 Literature

We know very little about geography of Colombian happiness–there are only 4 studies on the topic as far as we can tell: Peiró-Palomino et al. (2020)/Peiro-Palomino et al. (2021), PNUD (2023), Burger et al. (2021), and Okulicz-Kozaryn et al. (2026).

Peiró-Palomino et al. (2020)/Peiro-Palomino et al. (2021)[2] is a pioneering study here, but it uses an index of objective indicators of quality of life such as income, housing and pollution. There is no subjective wellbeing studied here. Still, their study has a useful information for our study, especially the map in their Figure 1: Global well-being in the Colombian departments, 2016. Their objective wellbeing findings are quite similar to our subjective wellbeing findings in next sections.

Burger et al. (2021) does analyze subjective wellbeing geography within Colombia, but has few survey observations: only several thousand (7-9k depending on model), and problematic measurement across space–it uses questionable Gallup data (see discussion in Okulicz-Kozaryn and Valente 2021), hence, we leave it out. Okulicz-Kozaryn et al. (2026) only focuses or urban-rural. While it also breaks down the analyses by municipalities, it only addresses Bogota, and lumps all the other municipalities into 2 categories (and like Burger et al. (2021) suffers from small sample).

Finally, the most comprehensive study to date: PNUD (2023) is a recent United Nations study on Colombian happiness using the same Encuesta Nacional de Calidad de Vida that we use, just earlier waves. The study is available from the the UN.[3] It is closest to our study and most useful overview of Colombian happiness geography to date. There is in fact a similar map to ours in Figure 1.9 on page 26 for 2018-2021. While happy Cafetero agrees with our analysis, happy Narino does not, and it finds Guaviare to be unhappy, while we find it moderately happy. A possible explanation for the differences is different time-frame: their 2018-2021 v our 2023. Likewise in figure 1.10 they break down SWB by urban-rural, our results agree that there is almost no difference, but we elaborate the relationship here: in addition to urban-rural we add farmer-nonfarmer and income breakdowns. PNUD (2023) also does analysis for municipalities in figures 2.14, 2.17 and 2.21 but the analyses do not address SWB, and only cover 10 cities, while we cover >40.

Overall, while there are at least 4 studies on the topic, only one (PNUD 2023) really has started addressing the life satisfaction geography of Colombia. These are, however, only rudimentary analyses among many, and ours is really the first study on the geography of Colombian life satisfaction.

[2] Both are the same study.

[3] https://www.undp.org/sites/g/files/zskgke326/files/2023-02/undp_co_pub_Percepciones_bienestar_subjetivo_2do_cuaderno_INDH_Feb21_2023.pdf.

2 Data

We use a fantastic datasource: 2023 Encuesta Nacional de Calidad de Vida (ECV). Sampling details are at `https://microdatos.dane.gov.co/index.php/catalog/734`

The ECV is nationally representative, cross-sectional annual survey in Colombia, it includes about 250k persons and 80k households across all 32 Colombian departments, rural-urban and many municipalities.[4]

The ECV identifies municipalities, but it is representative of departments only, not municipalities,[5] hence municipalities results are provisional and should be interpreted with care. Still, the samples are sizable and we are able to use >40 municipalities with at least 200 observations, and >30 with at least few hundred observations. In addition to 2023 ECV, we provide 2021 ECV robustness/sensitivity check in appendix.

`'Life satisfaction'` is measured with P1895 'Overall, how satisfied do you feel ... with your life at present?' on scale from 0='Completely dissatisfied' to 10='Completely satisfied.' The actual question asked in Spanish was: 'En general, qué tan satisfecho(a) se siente ... con su vida actualmente?' on scale from 0='Totalmente insatisfecho(a)' to 10='Totalmente satisfecho(a)'

One problem, unfortunately, for otherwise superb dataset, is that ECV has many missing values. On life satisfaction there are about 55k missing observations, another reason for robustness/sensitivity check in appendix using 2021 wave.

3 Departments/Provinces

We map `life satisfaction` by department in figure 1. For map readability we have not put names on departments, and instead we describe their locations below in text. For a map with department names see `https://www.google.com/search?q=colombia+departments&tbm=isch` or appendix section 'Administrative Boundaries.'

There are quite large differences close to 1 on 0-10 life satisfaction scale. The table including number of observations per province is in appendix. Again, in the body of the paper are results for 2023, appendix has robustness/sensitivity check for 2021.

[4] `https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-ecv-2021`

[5] See `https://microdatos.dane.gov.co/index.php/catalog/905` and `https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-ecv-2024`.

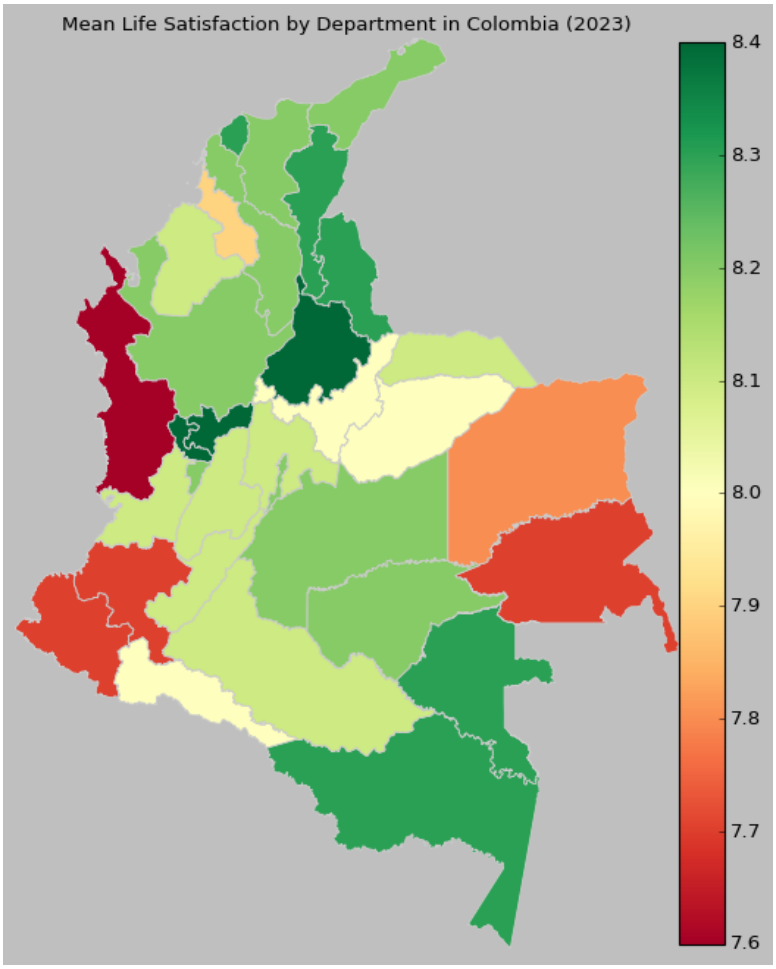


Figure 1

Unhappy departments are Western, South-Western, and Eastern periphery: poor Western Choco (dark cherry red), South-Western Cauca and most South-Western Narino (red); and on Eastern periphery Guainia (red) and Vichada (light red). In 2021 (see appendix) Choco and Cauca were still relatively unhappy, but slightly happier.

Happy departments are in central Colombia.[6] Cafetero (dark green East of cherry red Cauca): Caldas, Risaralda, Quindio–in 2021 there were slight differences (see appendix) but overall Cafetero has been happy, too. And also Santander is happy (big dark green North-East of Cafetero). Then slightly less happy Norte de Santander, Cesar and Atlantico (green North of Santander). In 2021 (see appendix) Santander, Cesar and Atlantico were less happy. The Southern-most Amazonas (green) is happy and so is Vaupes (green north of Amazonas).[7]

Caribbean (Northern coast) Cordoba (very light green) and Sucre (orange) are relatively unhappy, but then Bolivar (green) where touristy Cartagena is located is quite happy and even more so is Atlantico (small darker green North-East of Bolivar) where is Barranquilla, at least in 2023 (not so much in 2021).

Overall, figure 1 2023 results are similar to 2021 robustness/sensitivity check in appendix except: small sample Amazonas and Vaupés, both South-Eastern at 8.3 in 2023 v 7.6 in 2021, a .7 difference. Second largest differences are for Santander and Putumayo that have increased by .4 from 2021 to 2023.

[6] Technically this center is slightly to West, but as half of the country to the East is largely uninhabited pristine nature, the center of Colombia in everyday language is shifted to the West.

[7] Amazonas' and Vaupes' high happiness should be interpreted with caution as in 2021 they were 0.7 less happy at unhappy score of 7.6 (see appendix).

4 Municipalities

Next we turn to municipalities in figure 2. Appendix includes table with municipalities names, life satisfaction values and number of non-missing observations on life satisfaction.[8] Here the differences even larger than across provinces at little over 1 on 0-10 life satisfaction scale, as expected–smaller geographies have more variability,

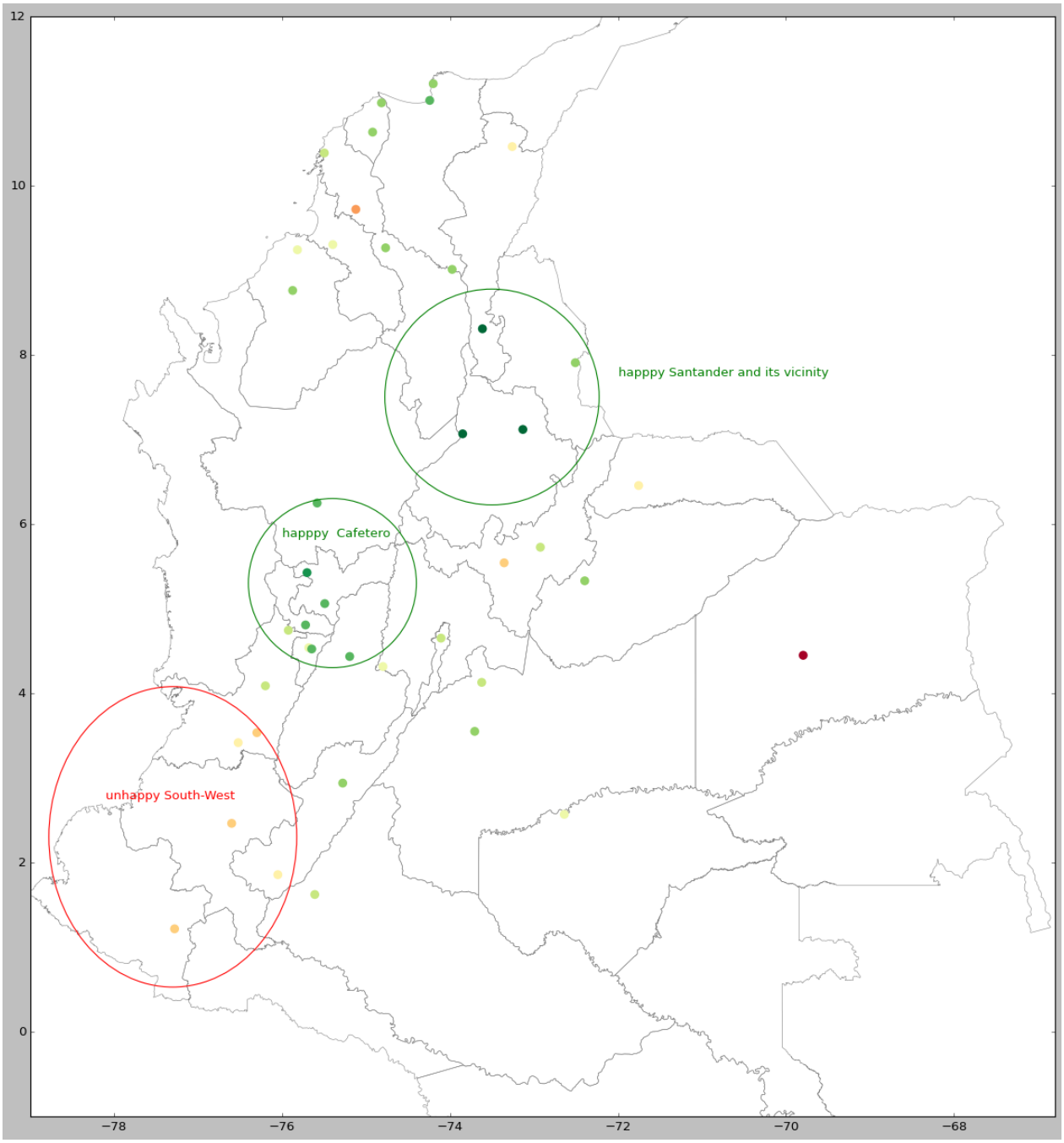


Figure 2: Life satisfaction across municipalities. Hi-res image–zoom in to see the detail. Cities with >100 non-missing life satisfaction observations per city shown. See appendix for details.

Santander is one of the happiest departments, and so 2 Santander largest cities are happy: Bucaramanga and Barrancabermeja at

[8]Note that many scores are different between 2021 and 2023–for many municipalities sample sizes are only few hundred and thus less stable. And again keep in mind that ECV is only representative of departments, not municipalities and hence municipalities results should be treated as provisional. Still, most municipalities results agree with departments–(un)happy municipalities tend to be in (un)happy departments.

8.6. Bucaramanga was less happy in 2021 at 8.1. Another happy region Cafetero has many happy cities: Calarcá, Pereira, Manizales at 8.4 and Armenia at 8.1 (Armenia was happier in 2021 at 8.5). Other happy cities include 2nd largest Medellin, Ibague, and Ciénaga at 8.4 (Ciénaga was unhappy in 2021 at 7.9). The least happy is remote Western Cumaribo at 7.5, and several cities at 7.9: Tunja, and Southern cities: Palmira in Valle de Cauca, Popayan in Cauca, and Pasto in Narino. In 2021 appendix results are similar, except that Santander and its vicinity is less happy. In the interpretation we have only focused on larger cities where sample size >200. Also see appendix for 2021-2023 differences.

Another limitation for municipalities is that we are missing smaller municipalities as they have smaller samples. We had to exclude smaller municipalities from the analysis as the sample sizes were too small. Hence, the results in this section do not generalize to smaller municipalities.

5 Urban v Rural and Farmers v Non-farmers

Urban v rural and farmers v non-farmers[9] is an important part of geographies of happiness. While in most developed countries cities are unhappy and indeed unhappy city is common across the world (Okulicz-Kozaryn and Valente 2021), it is not always the case in developing countries.

Cities are economic growth engines (Glaeser 2011, O'Sullivan 2009) and developing countries like Colombia do need economic growth–a role for cities is greater in developing countries than in already developed countries. And there is a "paradox"[10] of "happy peasant and unhappy millionaire" (Graham 2009). In Colombia, we only find conditional evidence for the paradox of happy peasant and unhappy millionaire–while urban and rural are about the same happy, there are differences by income categories.

`'Farmer'` is measured with p2057 'Do you consider yourself a farmer?' (¿Usted se considera campesino(a)? 1=Si 0=No) `'Rural'` is measured with P2061 'Do you consider the community where you live to be a rural farming community?' (¿Usted considera que la comunidad en que vive es campesina? 1=Si 0=No)

We plot life satisfaction for `rural` and `farmer` within each income[11] decile in figure 3–there are slight differences, but overall similar pattern in both panels. There is no difference for the bottom and 9th deciles. The bottom decile parity could be explained as that there is some minimum income necessary, and below it, it doesn't help to be a farmer or live in rural community.

`Rural` and `farmer` are happier in 2nd-8th deciles, especially in 2nd-6th deciles, and not rural and not farmer happier in the top decile. This makes sense, one needs money to enjoy city, where cost of living is higher. In other words, life is more commodified in the city–one needs to pay for more goods and services. For instance, many in the rural areas grow their own food (García-Quero and Guardiola 2018). In other words, there are lower returns in happiness from income for rural (farmers) v urban (non-farmers), or there is a bigger premium from income in urban.

One explanation could be that money can buy more varied and luxury goods and services in urban than rural, and one can enjoy more material comfort and consumption. Urbanity also provides more anonymity and freedom (Okulicz-Kozaryn 2015) and perhaps in such conditions more money can translate into more happiness. Another explanation is that non-farm and urban areas have higher cost of living. Likewise, in rural areas many goods and services are outside of the official economy. For instance, many grow their own food, have hens that lay eggs, use freely available materials such as wood, stones, and sand for construction, and so forth. Households that grow their own food and are in an indigenous community depend less on money to be happy (García-Quero and Guardiola 2018)

[9]The two typologies are very similar but not identical, there is some minimal urbal large city agriculture, and there is substantail farming in semi-urban small towns and large villages. But farming is mostly rural.

[10]Not so much paradox if you think about it–there is actually little reason for a millionaire to be happier than a peasant (Okulicz-Kozaryn and Martinez 2025).

[11]`'Income'` is 'per capita income' ('Ingreso per-capita'), i.e. total household income divided by the total number of household members. The sum of all monetary and non-monetary incomes of all household members. This includes wage salaries, independent earnings, retirement pensions, subsidies, and in-kind payments/goods. Per-Capita Equalization: By dividing this aggregated total by the count of residents under that roof, it normalizes income across households of varying sizes

Overall, it is worth emphasizing that except for the 1st decile and up to median and one decile above it (6th decile)–farmers and rural folks are clearly happier. For rural there are close to 0.2 on SWB, differences on 2nd and 3rd lowest bins: '128k-209k' and '210k-280k'.

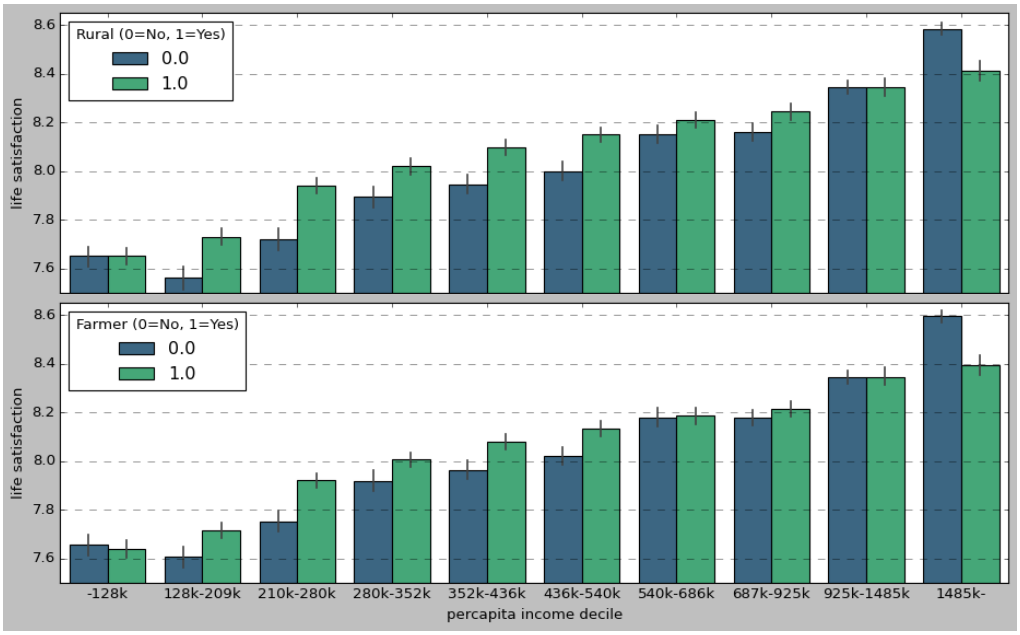


Figure 3

For `rural` in 2021 (appendix) pattern is similar but less strong for lower deciles where `rural` are happier than urban, but stronger for 2 top deciles where the opposite is true. For `farmer` in 2021 there is no happiness premium in lower deciles, but for non-farmers there is happiness premium in 2 top deciles.

6 Safety Satisfaction

Overall life satisfaction can be conceptualized as a sum of domain satisfactions (Campbell et al. 1976). We turn to safety, an important satisfaction domain in unsafe Colombia.

Threat of being a victim, fear of crime, and perception of unsafety are a part of Colombian wellbeing calculus. Colombia has been ruined by largest civil conflict in Latin America and drug-trafficking related violence in 80s and 90s (Martínez and Short 2020). Colombia is still haunted by violence and conflict, much of which is rural Turkewitz (2021).[12]

Crime rates are still high: Colombia ranks in top decile on homicide rate, and by one "criminality" index it ranks as the 2nd most "criminal" country in the world (after Myanmar; out of 193 countries).[13] Although it is arguably an overstatement in terms of how crime actually affects vast majority of Colombians–this "criminality" index weighs heavily organized crime networks that only affect an average citizen sporadically.

Violence decrease happiness, but less so when it has been prevalent for decades as it has been in Colombia–humans adapt to just about anything and it is difficult to overstate human adaptability (Graham 2009, Brickman et al. 1978).

Here we use subjective self-reported data from ECV (in appendix we also measure objective crime with homicide rate). `'Safety satisfaction'` is measured with P1898 'Overall, how satisfied do you feel ... with your current level of safety?' on scale from

[12] Also see `hrw.org/world-report/2020/country-chapters/colombia`.

[13] See `https://ocindex.net/report/2023/02-about-the-index.html` and 2/193 2023 criminality rank at `https://ocindex.net/rankings?f=rankings&view=List`.

0='Completely dissatisfied' to 10='Completely satisfied.' The actual question asked in Spanish was: 'En general, ¿qué tan satisfecho/a se siente ... con su nivel de seguridad actualmente?' on scale from 0='Totalmente insatisfecho(a)' to 10='Totalmente satisfecho(a)'

Figure 4 maps `safety satisfaction` in top panel, and in bottom panel `safety satisfaction` is correlated with `life satisfaction`.[14] As expected, the relationship is strong (and linear), and there are only handful of outliers marked in both panels.[15]

In Choco safety is quite low but not the very lowest at 7.1, but life satisfaction is the very lowest at 7.6, perhaps because Choco is the very poorest department in Colombia. Atlantico, on the other hand, is satisfied with safety only about average, but high on life satisfaction at 8.4–it may be Caribbean lifestyle, weather and music: very happy but not very safe. Interestingly both Choco and Atlantico are about the same safety satisfied but life satisfaction gap is great. The last outlier, Santander is only moderately satisfied with safety at 7.6 but is one of the very happiest at 8.4.

[14] Note that range of `safety satisfaction` is about 2x `life satisfaction`.
[15] Again, department map with department names on it is in appendix.

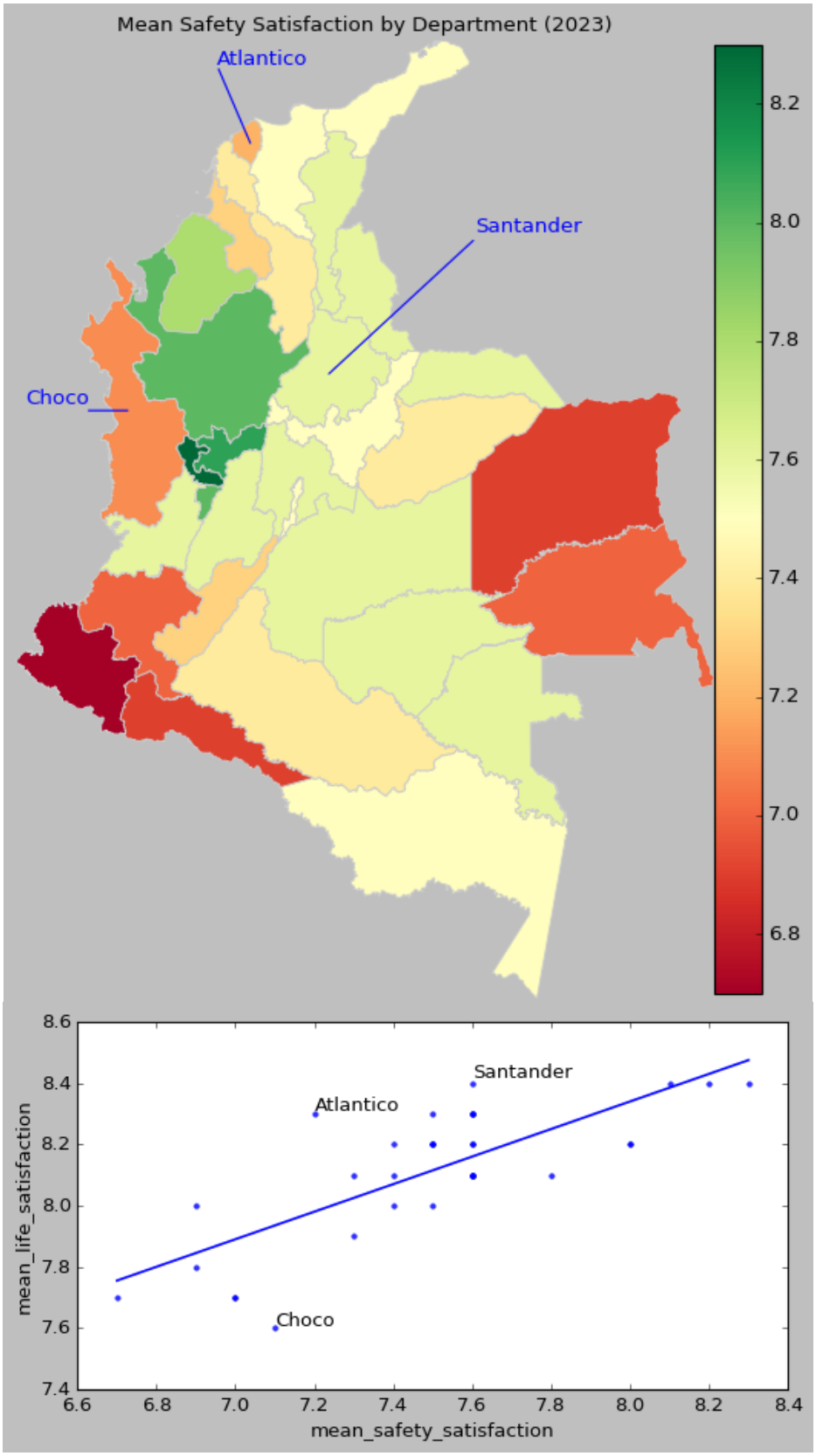

Mean Safety Satisfaction by Department (2023)
Atlantico
Santander
Choco
8.2
8.0
7.8
7.6
7.4
7.2
7.0
6.8
Santander
Atlantico
Choco
mean_life_satisfaction
mean_safety_satisfaction
8.6
8.4
8.2
8.0
7.8
7.6
7.4
6.6
6.8
7.0
7.2
7.4
7.6
7.8
8.0
8.2
8.4

Figure 4

7 Conclusion and Discussion

Colombia is a very diverse country in multiple dimensions: climate, topography, biomass and biodiversity, culture, ethnicity, economic development, and population density (see appendix). Here we have focused on differences in happiness across space.

Colombia, one of the very happiest countries in the world, has large happiness differences across geography within the country. Differences across the 32 departments are as large as almost 1 on 0-10 SWB scale. Differences across municipalities that were covered in the data are even larger at above 1 on 0-10 SWB scale. Farmers and non-urban settlements are happier than non-farmers and urbanity across income distribution except the 1st, 9th and 10th deciles. In deciles 2-6 the differences are as large as 0.2 on 0-10 SWB scale: farmers and rural communities are happier by about 0.2 than non farmers and urbanity. Thus we find support for 'happy peasant' (Graham 2009) and 'buen vivir' hypotheses (Guardiola and García-Quero 2014, García-Quero and Guardiola 2018)).

On the other hand, in last decile non-farmers and city dwellers are significantly happier–perhaps high income goes a longer way for non-farmers and cities. And in general urban life is more commodified–one has to pay for more goods and services than in rural areas. Related, non-farm and urban areas have a higher cost of living.

Finally, we find a strong and linear relationship between safety and life satisfactions. Safety is important in unsafe (and happy) Colombia.

We have used a fantastic and underused dataset, Encuesta Nacional de Calidad de Vida (ECV)–it has tens of thousands of observations and hundreds of variables. Colombia is the leader in SWB data collection.

This is the first study focusing on geography of Colombian happiness–the purpose here was to map the data and present basic relationships across space. Future research can analyze SWB by socio-demographic subgroups, say by age, education, and income–how different groups score on SWB across space. It is likely that younger, more educated and richer folks are happier in cities v rural areas. We have already found some evidence here with respect to income–for the richest income decile there was a happiness premium for non-farmers and cities.

There are limitations as alreaady hinted at throughout, about 55k out of about 250k missing observations on life satisfaction (and other variables have missing observations, too). Municipalities results are provisional as ECV is not representative of municipalities but only departments. Also we had to exclude smaller municipalities from the analysis as the sample sizes were too small, below 100. Still, municipalities results mostly agree with a robustness/sensitivity check using 2021 data un appendix, and with departments–(un)happy municipalities tend to be in (un)happy departments.

ONLINE APPENDIX (Supplementary Online Material (SOM)

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead]

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1 Significance testing

1.1 Rural and Farmer

1.1.1 Rural

```
. reg life_satisfaction i.rural##ib10.inc10, robust
Linear regression                               Number of obs     =    184,047
                                                F(19, 184027)     =     256.08
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0255
                                                Root MSE          =     1.6726
------------------------------------------------------------------------------
             |               Robust
life_satis~n | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     1.rural |      -0.16       0.02    -7.19    0.00        -0.21       -0.12
             |
       inc10 |
          1  |      -0.90       0.03   -34.58    0.00        -0.95       -0.85
          2  |      -0.97       0.03   -34.09    0.00        -1.03       -0.91
          3  |      -0.91       0.03   -32.62    0.00        -0.96       -0.85
          4  |      -0.75       0.03   -28.79    0.00        -0.80       -0.70
          5  |      -0.61       0.02   -25.43    0.00        -0.66       -0.56
          6  |      -0.60       0.02   -27.19    0.00        -0.65       -0.56
          7  |      -0.47       0.02   -21.64    0.00        -0.51       -0.43
          8  |      -0.41       0.02   -20.60    0.00        -0.45       -0.37
          9  |      -0.26       0.02   -14.39    0.00        -0.30       -0.22
             |
 rural#inc10 |
       1  1  |       0.17       0.04     4.43    0.00         0.09        0.24
       1  2  |       0.27       0.04     6.83    0.00         0.19        0.34
       1  3  |       0.40       0.04    10.57    0.00         0.32        0.47
       1  4  |       0.35       0.04     9.65    0.00         0.28        0.42
       1  5  |       0.32       0.03     9.11    0.00         0.25        0.38
       1  6  |       0.32       0.03     9.70    0.00         0.26        0.39
       1  7  |       0.23       0.03     6.98    0.00         0.17        0.30
       1  8  |       0.27       0.03     8.34    0.00         0.20        0.33
       1  9  |       0.17       0.03     5.36    0.00         0.11        0.23
             |
       _cons |       8.56       0.01   739.21    0.00         8.54        8.58
------------------------------------------------------------------------------


BELOW JUST SELECTED CONTRASTS, ALL AFTER THAT
. margins rural, at(inc10=(1(1)10)) pwcompare(effects)

. margins rural, at(inc10=(1(1)10)) pwcompare(effects)
Pairwise comparisons of adjusted predictions          Number of obs = 184,047
Model VCE: Robust

Expression: Linear prediction, predict()
1._at:  inc10 =  1
2._at:  inc10 =  2
3._at:  inc10 =  3
4._at:  inc10 =  4
5._at:  inc10 =  5
6._at:  inc10 =  6
7._at:  inc10 =  7
8._at:  inc10 =  8
9._at:  inc10 =  9
10._at: inc10 = 10
------------------------------------------------------------------------------
                 |            Delta-method    Unadjusted           Unadjusted
                 |   Contrast   std. err.      t    P>|t|     [95% conf. interval]
-----------------+------------------------------------------------------------
        _at#rural |
( 1 1) vs ( 1 0)  |       0.00       0.03     0.13    0.90        -0.05        0.06
insig
( 2 1) vs ( 2 0)  |       0.10       0.03     3.26    0.00         0.04        0.17
sig
( 3 1) vs ( 3 0)  |       0.24       0.03     7.82    0.00         0.18        0.29
sig
( 4 1) vs ( 4 0)  |       0.19       0.03     6.61    0.00         0.13        0.24
sig
( 5 1) vs ( 5 0)  |       0.15       0.03     5.83    0.00         0.10        0.21
sig
( 6 1) vs ( 6 0)  |       0.16       0.02     6.55    0.00         0.11        0.21
sig
( 7 1) vs ( 7 0)  |       0.07       0.02     2.83    0.00         0.02        0.12
```

```
sig
( 8 1) vs ( 8 0)  |       0.10       0.02     4.59    0.00         0.06        0.15
sig
( 9 1) vs ( 9 0)  |       0.00       0.02     0.22    0.83        -0.04        0.05
insig
(10 1) vs (10 0)  |      -0.16       0.02    -7.19    0.00        -0.21       -0.12
sig



. margins rural, at(inc10=(1(1)10)) pwcompare(effects)
Pairwise comparisons of adjusted predictions          Number of obs = 184,047
Model VCE: Robust

Expression: Linear prediction, predict()
1._at:  inc10 =  1
2._at:  inc10 =  2
3._at:  inc10 =  3
4._at:  inc10 =  4
5._at:  inc10 =  5
6._at:  inc10 =  6
7._at:  inc10 =  7
8._at:  inc10 =  8
9._at:  inc10 =  9
10._at: inc10 = 10
------------------------------------------------------------------------------
                   |            Delta-method    Unadjusted           Unadjusted
                   |   Contrast   std. err.      t    P>|t|     [95% conf. interval]
-------------------+----------------------------------------------------------------
         _at#rural |
( 1 1) vs ( 1 0)  |       0.00       0.03     0.13    0.90        -0.05        0.06
( 2 0) vs ( 1 0)  |      -0.07       0.03    -2.03    0.04        -0.14       -0.00
( 2 1) vs ( 1 0)  |       0.03       0.03     1.10    0.27        -0.03        0.09
( 3 0) vs ( 1 0)  |      -0.01       0.03    -0.27    0.78        -0.08        0.06
( 3 1) vs ( 1 0)  |       0.23       0.03     7.96    0.00         0.17        0.28
( 4 0) vs ( 1 0)  |       0.15       0.03     4.59    0.00         0.09        0.22
( 4 1) vs ( 1 0)  |       0.34       0.03    11.96    0.00         0.28        0.39
( 5 0) vs ( 1 0)  |       0.29       0.03     9.15    0.00         0.23        0.35
( 5 1) vs ( 1 0)  |       0.44       0.03    15.68    0.00         0.39        0.50
( 6 0) vs ( 1 0)  |       0.30       0.03     9.83    0.00         0.24        0.35
( 6 1) vs ( 1 0)  |       0.45       0.03    16.37    0.00         0.40        0.51
( 7 0) vs ( 1 0)  |       0.43       0.03    14.56    0.00         0.37        0.49
( 7 1) vs ( 1 0)  |       0.50       0.03    17.76    0.00         0.44        0.55
( 8 0) vs ( 1 0)  |       0.49       0.03    17.21    0.00         0.43        0.54
( 8 1) vs ( 1 0)  |       0.59       0.03    21.12    0.00         0.54        0.65
( 9 0) vs ( 1 0)  |       0.64       0.03    23.56    0.00         0.59        0.69
( 9 1) vs ( 1 0)  |       0.64       0.03    22.57    0.00         0.59        0.70
(10 0) vs ( 1 0)  |       0.90       0.03    34.58    0.00         0.85        0.95
(10 1) vs ( 1 0)  |       0.74       0.03    24.27    0.00         0.68        0.80
( 2 0) vs ( 1 1)  |      -0.07       0.03    -2.32    0.02        -0.14       -0.01
( 2 1) vs ( 1 1)  |       0.03       0.03     1.09    0.28        -0.02        0.08
( 3 0) vs ( 1 1)  |      -0.01       0.03    -0.42    0.68        -0.08        0.05
( 3 1) vs ( 1 1)  |       0.22       0.02     8.92    0.00         0.17        0.27
( 4 0) vs ( 1 1)  |       0.15       0.03     4.91    0.00         0.09        0.21
( 4 1) vs ( 1 1)  |       0.33       0.02    13.51    0.00         0.29        0.38
( 5 0) vs ( 1 1)  |       0.28       0.03    10.02    0.00         0.23        0.34
( 5 1) vs ( 1 1)  |       0.44       0.02    17.77    0.00         0.39        0.49
( 6 0) vs ( 1 1)  |       0.29       0.03    10.88    0.00         0.24        0.34
( 6 1) vs ( 1 1)  |       0.45       0.02    18.63    0.00         0.40        0.50
( 7 0) vs ( 1 1)  |       0.43       0.03    16.26    0.00         0.38        0.48
( 7 1) vs ( 1 1)  |       0.50       0.02    20.15    0.00         0.45        0.54
( 8 0) vs ( 1 1)  |       0.48       0.02    19.46    0.00         0.44        0.53
( 8 1) vs ( 1 1)  |       0.59       0.02    24.00    0.00         0.54        0.64
( 9 0) vs ( 1 1)  |       0.64       0.02    27.08    0.00         0.59        0.68
( 9 1) vs ( 1 1)  |       0.64       0.03    25.53    0.00         0.59        0.69
(10 0) vs ( 1 1)  |       0.90       0.02    40.42    0.00         0.85        0.94
(10 1) vs ( 1 1)  |       0.73       0.03    27.02    0.00         0.68        0.79
( 2 1) vs ( 2 0)  |       0.10       0.03     3.26    0.00         0.04        0.17
( 3 0) vs ( 2 0)  |       0.06       0.04     1.69    0.09        -0.01        0.13
( 3 1) vs ( 2 0)  |       0.30       0.03     9.68    0.00         0.24        0.36
( 4 0) vs ( 2 0)  |       0.22       0.03     6.36    0.00         0.15        0.29
( 4 1) vs ( 2 0)  |       0.41       0.03    13.39    0.00         0.35        0.47
( 5 0) vs ( 2 0)  |       0.36       0.03    10.70    0.00         0.29        0.42
( 5 1) vs ( 2 0)  |       0.51       0.03    16.83    0.00         0.45        0.57
( 6 0) vs ( 2 0)  |       0.37       0.03    11.37    0.00         0.30        0.43
( 6 1) vs ( 2 0)  |       0.52       0.03    17.47    0.00         0.47        0.58
( 7 0) vs ( 2 0)  |       0.50       0.03    15.79    0.00         0.44        0.56
( 7 1) vs ( 2 0)  |       0.57       0.03    18.76    0.00         0.51        0.63
( 8 0) vs ( 2 0)  |       0.56       0.03    18.25    0.00         0.50        0.62
( 8 1) vs ( 2 0)  |       0.66       0.03    21.86    0.00         0.60        0.72
( 9 0) vs ( 2 0)  |       0.71       0.03    24.08    0.00         0.65        0.77
( 9 1) vs ( 2 0)  |       0.71       0.03    23.22    0.00         0.65        0.78
(10 0) vs ( 2 0)  |       0.97       0.03    34.09    0.00         0.91        1.03
(10 1) vs ( 2 0)  |       0.81       0.03    24.86    0.00         0.74        0.87
( 3 0) vs ( 2 1)  |      -0.04       0.03    -1.34    0.18        -0.10        0.02
( 3 1) vs ( 2 1)  |       0.19       0.02     7.97    0.00         0.15        0.24
( 4 0) vs ( 2 1)  |       0.12       0.03     4.03    0.00         0.06        0.18
( 4 1) vs ( 2 1)  |       0.31       0.02    12.68    0.00         0.26        0.35
( 5 0) vs ( 2 1)  |       0.26       0.03     9.19    0.00         0.20        0.31
( 5 1) vs ( 2 1)  |       0.41       0.02    17.04    0.00         0.36        0.46
( 6 0) vs ( 2 1)  |       0.26       0.03    10.04    0.00         0.21        0.31
( 6 1) vs ( 2 1)  |       0.42       0.02    17.93    0.00         0.38        0.47
( 7 0) vs ( 2 1)  |       0.40       0.03    15.52    0.00         0.35        0.45
( 7 1) vs ( 2 1)  |       0.47       0.02    19.49    0.00         0.42        0.51
( 8 0) vs ( 2 1)  |       0.46       0.02    18.79    0.00         0.41        0.50
( 8 1) vs ( 2 1)  |       0.56       0.02    23.44    0.00         0.51        0.61
( 9 0) vs ( 2 1)  |       0.61       0.02    26.62    0.00         0.56        0.65
( 9 1) vs ( 2 1)  |       0.61       0.02    25.01    0.00         0.56        0.66
(10 0) vs ( 2 1)  |       0.87       0.02    40.42    0.00         0.83        0.91
(10 1) vs ( 2 1)  |       0.70       0.03    26.52    0.00         0.65        0.76
( 3 1) vs ( 3 0)  |       0.24       0.03     7.82    0.00         0.18        0.29
( 4 0) vs ( 3 0)  |       0.16       0.03     4.66    0.00         0.09        0.23
( 4 1) vs ( 3 0)  |       0.35       0.03    11.59    0.00         0.29        0.41
( 5 0) vs ( 3 0)  |       0.30       0.03     9.00    0.00         0.23        0.36
( 5 1) vs ( 3 0)  |       0.45       0.03    15.09    0.00         0.39        0.51
```

(6 0) vs (3 0)	0.30	0.03	9.62	0.00	0.24	0.37
(6 1) vs (3 0)	0.46	0.03	15.72	0.00	0.41	0.52
(7 0) vs (3 0)	0.44	0.03	14.10	0.00	0.38	0.50
(7 1) vs (3 0)	0.51	0.03	17.05	0.00	0.45	0.57
(8 0) vs (3 0)	0.50	0.03	16.55	0.00	0.44	0.56
(8 1) vs (3 0)	0.60	0.03	20.21	0.00	0.54	0.66
(9 0) vs (3 0)	0.65	0.03	22.44	0.00	0.59	0.71
(9 1) vs (3 0)	0.65	0.03	21.61	0.00	0.59	0.71
(10 0) vs (3 0)	0.91	0.03	32.62	0.00	0.85	0.96
(10 1) vs (3 0)	0.75	0.03	23.35	0.00	0.68	0.81
(4 0) vs (3 1)	-0.07	0.03	-2.64	0.01	-0.13	-0.02
(4 1) vs (3 1)	0.11	0.02	4.91	0.00	0.07	0.16
(5 0) vs (3 1)	0.06	0.03	2.32	0.02	0.01	0.11
(5 1) vs (3 1)	0.22	0.02	9.52	0.00	0.17	0.26
(6 0) vs (3 1)	0.07	0.02	2.78	0.01	0.02	0.12
(6 1) vs (3 1)	0.23	0.02	10.32	0.00	0.18	0.27
(7 0) vs (3 1)	0.21	0.02	8.41	0.00	0.16	0.25
(7 1) vs (3 1)	0.27	0.02	12.11	0.00	0.23	0.32
(8 0) vs (3 1)	0.26	0.02	11.46	0.00	0.22	0.31
(8 1) vs (3 1)	0.37	0.02	16.28	0.00	0.32	0.41
(9 0) vs (3 1)	0.41	0.02	19.36	0.00	0.37	0.46
(9 1) vs (3 1)	0.42	0.02	18.09	0.00	0.37	0.46
(10 0) vs (3 1)	0.67	0.02	33.83	0.00	0.63	0.71
(10 1) vs (3 1)	0.51	0.03	20.17	0.00	0.46	0.56
(4 1) vs (4 0)	0.19	0.03	6.61	0.00	0.13	0.24
(5 0) vs (4 0)	0.14	0.03	4.34	0.00	0.07	0.20
(5 1) vs (4 0)	0.29	0.03	10.31	0.00	0.23	0.35
(6 0) vs (4 0)	0.14	0.03	4.80	0.00	0.09	0.20
(6 1) vs (4 0)	0.30	0.03	10.93	0.00	0.25	0.36
(7 0) vs (4 0)	0.28	0.03	9.46	0.00	0.22	0.34
(7 1) vs (4 0)	0.35	0.03	12.39	0.00	0.29	0.40
(8 0) vs (4 0)	0.34	0.03	11.89	0.00	0.28	0.39
(8 1) vs (4 0)	0.44	0.03	15.74	0.00	0.39	0.50
(9 0) vs (4 0)	0.49	0.03	18.00	0.00	0.44	0.54
(9 1) vs (4 0)	0.49	0.03	17.28	0.00	0.44	0.55
(10 0) vs (4 0)	0.75	0.03	28.79	0.00	0.70	0.80
(10 1) vs (4 0)	0.59	0.03	19.30	0.00	0.53	0.65
(5 0) vs (4 1)	-0.05	0.03	-1.88	0.06	-0.10	0.00
(5 1) vs (4 1)	0.10	0.02	4.64	0.00	0.06	0.15
(6 0) vs (4 1)	-0.04	0.02	-1.70	0.09	-0.09	0.01
(6 1) vs (4 1)	0.12	0.02	5.34	0.00	0.07	0.16
(7 0) vs (4 1)	0.09	0.02	3.88	0.00	0.05	0.14
(7 1) vs (4 1)	0.16	0.02	7.25	0.00	0.12	0.21
(8 0) vs (4 1)	0.15	0.02	6.67	0.00	0.11	0.20
(8 1) vs (4 1)	0.25	0.02	11.45	0.00	0.21	0.30
(9 0) vs (4 1)	0.30	0.02	14.31	0.00	0.26	0.34
(9 1) vs (4 1)	0.31	0.02	13.40	0.00	0.26	0.35
(10 0) vs (4 1)	0.56	0.02	28.61	0.00	0.52	0.60
(10 1) vs (4 1)	0.40	0.03	15.90	0.00	0.35	0.45
(5 1) vs (5 0)	0.15	0.03	5.83	0.00	0.10	0.21
(6 0) vs (5 0)	0.01	0.03	0.27	0.78	-0.05	0.06
(6 1) vs (5 0)	0.17	0.03	6.42	0.00	0.12	0.22
(7 0) vs (5 0)	0.14	0.03	5.15	0.00	0.09	0.20
(7 1) vs (5 0)	0.21	0.03	8.05	0.00	0.16	0.26
(8 0) vs (5 0)	0.20	0.03	7.56	0.00	0.15	0.25
(8 1) vs (5 0)	0.30	0.03	11.61	0.00	0.25	0.36
(9 0) vs (5 0)	0.35	0.03	13.92	0.00	0.30	0.40
(9 1) vs (5 0)	0.36	0.03	13.31	0.00	0.30	0.41
(10 0) vs (5 0)	0.61	0.02	25.43	0.00	0.56	0.66
(10 1) vs (5 0)	0.45	0.03	15.65	0.00	0.39	0.51
(6 0) vs (5 1)	-0.15	0.02	-5.91	0.00	-0.19	-0.10
(6 1) vs (5 1)	0.01	0.02	0.59	0.55	-0.03	0.06
(7 0) vs (5 1)	-0.01	0.02	-0.41	0.68	-0.06	0.04
(7 1) vs (5 1)	0.06	0.02	2.62	0.01	0.01	0.10
(8 0) vs (5 1)	0.05	0.02	2.10	0.04	0.00	0.09
(8 1) vs (5 1)	0.15	0.02	6.80	0.00	0.11	0.19
(9 0) vs (5 1)	0.20	0.02	9.42	0.00	0.16	0.24
(9 1) vs (5 1)	0.20	0.02	8.89	0.00	0.16	0.25
(10 0) vs (5 1)	0.46	0.02	23.42	0.00	0.42	0.50
(10 1) vs (5 1)	0.30	0.03	11.80	0.00	0.25	0.34
(6 1) vs (6 0)	0.16	0.02	6.55	0.00	0.11	0.21
(7 0) vs (6 0)	0.14	0.03	5.16	0.00	0.08	0.19
(7 1) vs (6 0)	0.20	0.02	8.28	0.00	0.16	0.25
(8 0) vs (6 0)	0.19	0.02	7.74	0.00	0.14	0.24
(8 1) vs (6 0)	0.30	0.02	12.08	0.00	0.25	0.35
(9 0) vs (6 0)	0.34	0.02	14.62	0.00	0.30	0.39
(9 1) vs (6 0)	0.35	0.03	13.87	0.00	0.30	0.40
(10 0) vs (6 0)	0.60	0.02	27.19	0.00	0.56	0.65
(10 1) vs (6 0)	0.44	0.03	16.24	0.00	0.39	0.49
(7 0) vs (6 1)	-0.02	0.02	-0.96	0.34	-0.07	0.02
(7 1) vs (6 1)	0.05	0.02	2.08	0.04	0.00	0.09
(8 0) vs (6 1)	0.03	0.02	1.56	0.12	-0.01	0.08
(8 1) vs (6 1)	0.14	0.02	6.36	0.00	0.10	0.18
(9 0) vs (6 1)	0.19	0.02	9.03	0.00	0.15	0.23
(9 1) vs (6 1)	0.19	0.02	8.50	0.00	0.15	0.23
(10 0) vs (6 1)	0.45	0.02	23.45	0.00	0.41	0.48
(10 1) vs (6 1)	0.28	0.02	11.48	0.00	0.23	0.33
(7 1) vs (7 0)	0.07	0.02	2.83	0.00	0.02	0.12
(8 0) vs (7 0)	0.06	0.02	2.35	0.02	0.01	0.11
(8 1) vs (7 0)	0.16	0.02	6.69	0.00	0.11	0.21
(9 0) vs (7 0)	0.21	0.02	9.05	0.00	0.16	0.25
(9 1) vs (7 0)	0.21	0.02	8.64	0.00	0.16	0.26
(10 0) vs (7 0)	0.47	0.02	21.64	0.00	0.43	0.51
(10 1) vs (7 0)	0.31	0.03	11.44	0.00	0.25	0.36
(8 0) vs (7 1)	-0.01	0.02	-0.48	0.63	-0.06	0.03
(8 1) vs (7 1)	0.09	0.02	4.18	0.00	0.05	0.14
(9 0) vs (7 1)	0.14	0.02	6.65	0.00	0.10	0.18
(9 1) vs (7 1)	0.14	0.02	6.34	0.00	0.10	0.19
(10 0) vs (7 1)	0.40	0.02	20.47	0.00	0.36	0.44
(10 1) vs (7 1)	0.24	0.03	9.47	0.00	0.19	0.29
(8 1) vs (8 0)	0.10	0.02	4.59	0.00	0.06	0.15
(9 0) vs (8 0)	0.15	0.02	7.04	0.00	0.11	0.19
(9 1) vs (8 0)	0.16	0.02	6.71	0.00	0.11	0.20
(10 0) vs (8 0)	0.41	0.02	20.60	0.00	0.37	0.45
(10 1) vs (8 0)	0.25	0.03	9.78	0.00	0.20	0.30
(9 0) vs (8 1)	0.05	0.02	2.25	0.02	0.01	0.09
(9 1) vs (8 1)	0.05	0.02	2.28	0.02	0.01	0.10

```
(10 0) vs ( 8 1)  |       0.31       0.02    15.81    0.00         0.27        0.35
(10 1) vs ( 8 1)  |       0.14       0.02     5.79    0.00         0.10        0.19
( 9 1) vs ( 9 0)  |       0.00       0.02     0.22    0.83        -0.04        0.05
(10 0) vs ( 9 0)  |       0.26       0.02    14.39    0.00         0.22        0.30
(10 1) vs ( 9 0)  |       0.10       0.02     4.07    0.00         0.05        0.14
(10 0) vs ( 9 1)  |       0.26       0.02    12.69    0.00         0.22        0.30
(10 1) vs ( 9 1)  |       0.09       0.03     3.64    0.00         0.04        0.14
(10 1) vs (10 0)  |      -0.16       0.02    -7.19    0.00        -0.21       -0.12
------------------------------------------------------------------------------
```

1.1.2 Farmer

```
. reg life_satisfaction i.farmer#ib10.inc10, robust
Linear regression                               Number of obs     =    185,236
                                                F(19, 185216)     =     255.75
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0250
                                                Root MSE          =     1.6737
------------------------------------------------------------------------------
             |               Robust
life_satis~n | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
farmer#inc10 |
       0  1  |      -0.90       0.03   -34.92    0.00        -0.95       -0.85
       0  2  |      -0.95       0.03   -34.41    0.00        -1.00       -0.89
       0  3  |      -0.87       0.03   -32.33    0.00        -0.93       -0.82
       0  4  |      -0.72       0.03   -28.14    0.00        -0.77       -0.67
       0  5  |      -0.60       0.02   -25.34    0.00        -0.65       -0.55
       0  6  |      -0.57       0.02   -25.88    0.00        -0.62       -0.53
       0  7  |      -0.46       0.02   -21.40    0.00        -0.50       -0.42
       0  8  |      -0.39       0.02   -19.73    0.00        -0.43       -0.35
       0  9  |      -0.26       0.02   -14.27    0.00        -0.29       -0.22
       1  1  |      -0.91       0.02   -41.12    0.00        -0.96       -0.87
       1  2  |      -0.88       0.02   -40.86    0.00        -0.93       -0.84
       1  3  |      -0.70       0.02   -34.70    0.00        -0.73       -0.66
       1  4  |      -0.58       0.02   -29.65    0.00        -0.62       -0.55
       1  5  |      -0.47       0.02   -24.14    0.00        -0.51       -0.44
       1  6  |      -0.48       0.02   -25.30    0.00        -0.52       -0.44
       1  7  |      -0.42       0.02   -21.32    0.00        -0.46       -0.38
       1  8  |      -0.34       0.02   -17.49    0.00        -0.38       -0.30
       1  9  |      -0.27       0.02   -13.48    0.00        -0.31       -0.23
       1 10  |      -0.17       0.02    -7.59    0.00        -0.22       -0.13
             |
       _cons |       8.56       0.01   745.87    0.00         8.54        8.59
------------------------------------------------------------------------------


BELOW JUST SELECTED CONTRASTS, ALL AFTER THAT
margins farmer, at(inc10=(1(1)10)) pwcompare(effects)
Pairwise comparisons of adjusted predictions          Number of obs = 185,236
Model VCE: Robust
Expression: Linear prediction, predict()
1._at:  inc10 =  1
2._at:  inc10 =  2
3._at:  inc10 =  3
4._at:  inc10 =  4
5._at:  inc10 =  5
6._at:  inc10 =  6
7._at:  inc10 =  7
8._at:  inc10 =  8
9._at:  inc10 =  9
10._at: inc10 = 10
------------------------------------------------------------------------------
                 |            Delta-method    Unadjusted           Unadjusted
                 |   Contrast   std. err.      t    P>|t|     [95% conf. interval]
-----------------+------------------------------------------------------------
       _at#farmer |
( 1 1) vs ( 1 0)  |      -0.02       0.03    -0.60    0.55        -0.08        0.04
insig
( 2 1) vs ( 2 0)  |       0.06       0.03     2.03    0.04         0.00        0.12
sig
( 3 1) vs ( 3 0)  |       0.18       0.03     6.07    0.00         0.12        0.24
sig
( 4 1) vs ( 4 0)  |       0.13       0.03     4.77    0.00         0.08        0.19
sig
( 5 1) vs ( 5 0)  |       0.13       0.03     4.86    0.00         0.08        0.18
sig
( 6 1) vs ( 6 0)  |       0.10       0.02     3.98    0.00         0.05        0.14
sig
( 7 1) vs ( 7 0)  |       0.04       0.02     1.71    0.09        -0.01        0.09
insig; only sig at .1
( 8 1) vs ( 8 0)  |       0.05       0.02     2.41    0.02         0.01        0.10
sig
( 9 1) vs ( 9 0)  |      -0.01       0.02    -0.48    0.63        -0.05        0.03
insig
(10 1) vs (10 0)  |      -0.17       0.02    -7.59    0.00        -0.22       -0.13
sig



. margins farmer, at(inc10=(1(1)10)) pwcompare(effects)
```

```
margins farmer, at(inc10=(1(1)10)) pwcompare(effects)
Pairwise comparisons of adjusted predictions          Number of obs = 185,236
Model VCE: Robust

Expression: Linear prediction, predict()
1._at:  inc10 =  1
2._at:  inc10 =  2
3._at:  inc10 =  3
4._at:  inc10 =  4
5._at:  inc10 =  5
6._at:  inc10 =  6
7._at:  inc10 =  7
8._at:  inc10 =  8
9._at:  inc10 =  9
10._at: inc10 = 10
```

		Delta-method	Unadjusted		Unadjusted	
	Contrast	std. err.	t	P>|t|	[95% conf.	interval]
_at#farmer						
(1 1) vs (1 0)	-0.02	0.03	-0.60	0.55	-0.08	0.04
(2 0) vs (1 0)	-0.05	0.03	-1.49	0.14	-0.12	0.02
(2 1) vs (1 0)	0.01	0.03	0.42	0.68	-0.05	0.07
(3 0) vs (1 0)	0.02	0.03	0.65	0.51	-0.04	0.09
(3 1) vs (1 0)	0.20	0.03	7.13	0.00	0.15	0.26
(4 0) vs (1 0)	0.18	0.03	5.52	0.00	0.12	0.24
(4 1) vs (1 0)	0.31	0.03	11.12	0.00	0.26	0.37
(5 0) vs (1 0)	0.30	0.03	9.55	0.00	0.23	0.36
(5 1) vs (1 0)	0.42	0.03	15.12	0.00	0.37	0.48
(6 0) vs (1 0)	0.32	0.03	10.78	0.00	0.26	0.38
(6 1) vs (1 0)	0.42	0.03	15.23	0.00	0.36	0.47
(7 0) vs (1 0)	0.44	0.03	14.98	0.00	0.38	0.50
(7 1) vs (1 0)	0.48	0.03	17.16	0.00	0.42	0.53
(8 0) vs (1 0)	0.50	0.03	17.90	0.00	0.45	0.56
(8 1) vs (1 0)	0.56	0.03	20.11	0.00	0.50	0.61
(9 0) vs (1 0)	0.64	0.03	23.73	0.00	0.59	0.69
(9 1) vs (1 0)	0.63	0.03	22.30	0.00	0.57	0.68
(10 0) vs (1 0)	0.90	0.03	34.92	0.00	0.85	0.95
(10 1) vs (1 0)	0.72	0.03	24.05	0.00	0.67	0.78
(2 0) vs (1 1)	-0.03	0.03	-1.04	0.30	-0.09	0.03
(2 1) vs (1 1)	0.03	0.03	1.14	0.25	-0.02	0.08
(3 0) vs (1 1)	0.04	0.03	1.29	0.20	-0.02	0.10
(3 1) vs (1 1)	0.22	0.03	8.71	0.00	0.17	0.27
(4 0) vs (1 1)	0.20	0.03	6.62	0.00	0.14	0.25
(4 1) vs (1 1)	0.33	0.02	13.23	0.00	0.28	0.38
(5 0) vs (1 1)	0.31	0.03	11.13	0.00	0.26	0.37
(5 1) vs (1 1)	0.44	0.02	17.74	0.00	0.39	0.49
(6 0) vs (1 1)	0.34	0.03	12.61	0.00	0.29	0.39
(6 1) vs (1 1)	0.44	0.02	17.96	0.00	0.39	0.48
(7 0) vs (1 1)	0.46	0.03	17.35	0.00	0.40	0.51
(7 1) vs (1 1)	0.50	0.02	20.05	0.00	0.45	0.55
(8 0) vs (1 1)	0.52	0.03	20.83	0.00	0.47	0.57
(8 1) vs (1 1)	0.58	0.02	23.40	0.00	0.53	0.62
(9 0) vs (1 1)	0.66	0.02	27.76	0.00	0.61	0.70
(9 1) vs (1 1)	0.65	0.03	25.76	0.00	0.60	0.69
(10 0) vs (1 1)	0.91	0.02	41.12	0.00	0.87	0.96
(10 1) vs (1 1)	0.74	0.03	27.23	0.00	0.69	0.80
(2 1) vs (2 0)	0.06	0.03	2.03	0.04	0.00	0.12
(3 0) vs (2 0)	0.07	0.04	2.08	0.04	0.00	0.14
(3 1) vs (2 0)	0.25	0.03	8.42	0.00	0.19	0.31
(4 0) vs (2 0)	0.23	0.03	6.78	0.00	0.16	0.30
(4 1) vs (2 0)	0.36	0.03	12.19	0.00	0.30	0.42
(5 0) vs (2 0)	0.35	0.03	10.65	0.00	0.28	0.41
(5 1) vs (2 0)	0.47	0.03	15.96	0.00	0.42	0.53
(6 0) vs (2 0)	0.37	0.03	11.84	0.00	0.31	0.43
(6 1) vs (2 0)	0.47	0.03	16.06	0.00	0.41	0.53
(7 0) vs (2 0)	0.49	0.03	15.83	0.00	0.43	0.55
(7 1) vs (2 0)	0.53	0.03	17.88	0.00	0.47	0.59
(8 0) vs (2 0)	0.55	0.03	18.58	0.00	0.50	0.61
(8 1) vs (2 0)	0.61	0.03	20.65	0.00	0.55	0.67
(9 0) vs (2 0)	0.69	0.03	24.03	0.00	0.63	0.74
(9 1) vs (2 0)	0.68	0.03	22.72	0.00	0.62	0.74
(10 0) vs (2 0)	0.95	0.03	34.41	0.00	0.89	1.00
(10 1) vs (2 0)	0.78	0.03	24.44	0.00	0.71	0.84
(3 0) vs (2 1)	0.01	0.03	0.32	0.75	-0.05	0.07
(3 1) vs (2 1)	0.19	0.02	7.67	0.00	0.14	0.24
(4 0) vs (2 1)	0.17	0.03	5.69	0.00	0.11	0.22
(4 1) vs (2 1)	0.30	0.02	12.28	0.00	0.25	0.35
(5 0) vs (2 1)	0.28	0.03	10.23	0.00	0.23	0.34
(5 1) vs (2 1)	0.41	0.02	16.89	0.00	0.36	0.46
(6 0) vs (2 1)	0.31	0.03	11.70	0.00	0.26	0.36
(6 1) vs (2 1)	0.41	0.02	17.10	0.00	0.36	0.45
(7 0) vs (2 1)	0.43	0.03	16.52	0.00	0.38	0.48
(7 1) vs (2 1)	0.47	0.02	19.24	0.00	0.42	0.51
(8 0) vs (2 1)	0.49	0.02	20.05	0.00	0.44	0.54
(8 1) vs (2 1)	0.55	0.02	22.67	0.00	0.50	0.59
(9 0) vs (2 1)	0.63	0.02	27.12	0.00	0.58	0.67
(9 1) vs (2 1)	0.62	0.02	25.08	0.00	0.57	0.66
(10 0) vs (2 1)	0.88	0.02	40.86	0.00	0.84	0.93
(10 1) vs (2 1)	0.71	0.03	26.59	0.00	0.66	0.76
(3 1) vs (3 0)	0.18	0.03	6.07	0.00	0.12	0.24
(4 0) vs (3 0)	0.16	0.03	4.68	0.00	0.09	0.22
(4 1) vs (3 0)	0.29	0.03	9.89	0.00	0.23	0.35
(5 0) vs (3 0)	0.27	0.03	8.52	0.00	0.21	0.34
(5 1) vs (3 0)	0.40	0.03	13.71	0.00	0.34	0.46
(6 0) vs (3 0)	0.30	0.03	9.65	0.00	0.24	0.36
(6 1) vs (3 0)	0.40	0.03	13.77	0.00	0.34	0.45
(7 0) vs (3 0)	0.42	0.03	13.66	0.00	0.36	0.48
(7 1) vs (3 0)	0.46	0.03	15.66	0.00	0.40	0.51
(8 0) vs (3 0)	0.48	0.03	16.38	0.00	0.42	0.54
(8 1) vs (3 0)	0.54	0.03	18.46	0.00	0.48	0.59
(9 0) vs (3 0)	0.62	0.03	21.84	0.00	0.56	0.67
(9 1) vs (3 0)	0.61	0.03	20.59	0.00	0.55	0.66
(10 0) vs (3 0)	0.87	0.03	32.33	0.00	0.82	0.93
(10 1) vs (3 0)	0.70	0.03	22.43	0.00	0.64	0.76
(4 0) vs (3 1)	-0.02	0.03	-0.80	0.42	-0.08	0.03
(4 1) vs (3 1)	0.11	0.02	4.81	0.00	0.07	0.16
(5 0) vs (3 1)	0.09	0.03	3.57	0.00	0.04	0.15

(5 1) vs (3 1)	0.22	0.02	9.68	0.00	0.18	0.27
(6 0) vs (3 1)	0.12	0.03	4.79	0.00	0.07	0.17
(6 1) vs (3 1)	0.22	0.02	9.74	0.00	0.17	0.26
(7 0) vs (3 1)	0.24	0.02	9.69	0.00	0.19	0.28
(7 1) vs (3 1)	0.28	0.02	12.17	0.00	0.23	0.32
(8 0) vs (3 1)	0.30	0.02	13.09	0.00	0.26	0.35
(8 1) vs (3 1)	0.36	0.02	15.76	0.00	0.31	0.40
(9 0) vs (3 1)	0.44	0.02	20.25	0.00	0.39	0.48
(9 1) vs (3 1)	0.43	0.02	18.44	0.00	0.38	0.47
(10 0) vs (3 1)	0.70	0.02	34.70	0.00	0.66	0.73
(10 1) vs (3 1)	0.52	0.03	20.52	0.00	0.47	0.57
(4 1) vs (4 0)	0.13	0.03	4.77	0.00	0.08	0.19
(5 0) vs (4 0)	0.12	0.03	3.79	0.00	0.06	0.18
(5 1) vs (4 0)	0.24	0.03	8.77	0.00	0.19	0.30
(6 0) vs (4 0)	0.14	0.03	4.81	0.00	0.08	0.20
(6 1) vs (4 0)	0.24	0.03	8.77	0.00	0.19	0.29
(7 0) vs (4 0)	0.26	0.03	8.91	0.00	0.20	0.32
(7 1) vs (4 0)	0.30	0.03	10.82	0.00	0.25	0.35
(8 0) vs (4 0)	0.33	0.03	11.62	0.00	0.27	0.38
(8 1) vs (4 0)	0.38	0.03	13.74	0.00	0.33	0.43
(9 0) vs (4 0)	0.46	0.03	17.18	0.00	0.41	0.51
(9 1) vs (4 0)	0.45	0.03	16.03	0.00	0.39	0.50
(10 0) vs (4 0)	0.72	0.03	28.14	0.00	0.67	0.77
(10 1) vs (4 0)	0.55	0.03	18.20	0.00	0.49	0.60
(5 0) vs (4 1)	-0.02	0.03	-0.61	0.54	-0.07	0.04
(5 1) vs (4 1)	0.11	0.02	4.91	0.00	0.07	0.16
(6 0) vs (4 1)	0.01	0.02	0.40	0.69	-0.04	0.06
(6 1) vs (4 1)	0.11	0.02	4.84	0.00	0.06	0.15
(7 0) vs (4 1)	0.13	0.02	5.23	0.00	0.08	0.17
(7 1) vs (4 1)	0.17	0.02	7.42	0.00	0.12	0.21
(8 0) vs (4 1)	0.19	0.02	8.42	0.00	0.15	0.24
(8 1) vs (4 1)	0.25	0.02	11.02	0.00	0.20	0.29
(9 0) vs (4 1)	0.33	0.02	15.34	0.00	0.28	0.37
(9 1) vs (4 1)	0.32	0.02	13.83	0.00	0.27	0.36
(10 0) vs (4 1)	0.58	0.02	29.65	0.00	0.55	0.62
(10 1) vs (4 1)	0.41	0.03	16.35	0.00	0.36	0.46
(5 1) vs (5 0)	0.13	0.03	4.86	0.00	0.08	0.18
(6 0) vs (5 0)	0.03	0.03	0.92	0.36	-0.03	0.08
(6 1) vs (5 0)	0.12	0.03	4.78	0.00	0.07	0.17
(7 0) vs (5 0)	0.14	0.03	5.17	0.00	0.09	0.20
(7 1) vs (5 0)	0.18	0.03	7.02	0.00	0.13	0.23
(8 0) vs (5 0)	0.21	0.03	7.90	0.00	0.16	0.26
(8 1) vs (5 0)	0.26	0.03	10.11	0.00	0.21	0.31
(9 0) vs (5 0)	0.34	0.03	13.69	0.00	0.29	0.39
(9 1) vs (5 0)	0.33	0.03	12.59	0.00	0.28	0.38
(10 0) vs (5 0)	0.60	0.02	25.34	0.00	0.55	0.65
(10 1) vs (5 0)	0.43	0.03	15.06	0.00	0.37	0.48
(6 0) vs (5 1)	-0.10	0.02	-4.07	0.00	-0.15	-0.05
(6 1) vs (5 1)	-0.00	0.02	-0.21	0.83	-0.05	0.04
(7 0) vs (5 1)	0.02	0.02	0.64	0.52	-0.03	0.06
(7 1) vs (5 1)	0.06	0.02	2.51	0.01	0.01	0.10
(8 0) vs (5 1)	0.08	0.02	3.57	0.00	0.04	0.13
(8 1) vs (5 1)	0.14	0.02	6.08	0.00	0.09	0.18
(9 0) vs (5 1)	0.22	0.02	10.17	0.00	0.17	0.26
(9 1) vs (5 1)	0.21	0.02	9.00	0.00	0.16	0.25
(10 0) vs (5 1)	0.47	0.02	24.14	0.00	0.44	0.51
(10 1) vs (5 1)	0.30	0.03	11.99	0.00	0.25	0.35
(6 1) vs (6 0)	0.10	0.02	3.98	0.00	0.05	0.14
(7 0) vs (6 0)	0.12	0.03	4.43	0.00	0.06	0.17
(7 1) vs (6 0)	0.16	0.02	6.36	0.00	0.11	0.21
(8 0) vs (6 0)	0.18	0.03	7.28	0.00	0.13	0.23
(8 1) vs (6 0)	0.24	0.02	9.62	0.00	0.19	0.28
(9 0) vs (6 0)	0.32	0.02	13.41	0.00	0.27	0.36
(9 1) vs (6 0)	0.31	0.03	12.23	0.00	0.26	0.36
(10 0) vs (6 0)	0.57	0.02	25.88	0.00	0.53	0.62
(10 1) vs (6 0)	0.40	0.03	14.79	0.00	0.35	0.46
(7 0) vs (6 1)	0.02	0.02	0.85	0.40	-0.03	0.07
(7 1) vs (6 1)	0.06	0.02	2.79	0.01	0.02	0.10
(8 0) vs (6 1)	0.09	0.02	3.88	0.00	0.04	0.13
(8 1) vs (6 1)	0.14	0.02	6.47	0.00	0.10	0.18
(9 0) vs (6 1)	0.22	0.02	10.72	0.00	0.18	0.26
(9 1) vs (6 1)	0.21	0.02	9.46	0.00	0.17	0.25
(10 0) vs (6 1)	0.48	0.02	25.30	0.00	0.44	0.52
(10 1) vs (6 1)	0.31	0.02	12.45	0.00	0.26	0.35
(7 1) vs (7 0)	0.04	0.02	1.71	0.09	-0.01	0.09
(8 0) vs (7 0)	0.07	0.02	2.71	0.01	0.02	0.11
(8 1) vs (7 0)	0.12	0.02	5.03	0.00	0.07	0.17
(9 0) vs (7 0)	0.20	0.02	8.75	0.00	0.16	0.24
(9 1) vs (7 0)	0.19	0.02	7.80	0.00	0.14	0.24
(10 0) vs (7 0)	0.46	0.02	21.40	0.00	0.42	0.50
(10 1) vs (7 0)	0.29	0.03	10.77	0.00	0.23	0.34
(8 0) vs (7 1)	0.02	0.02	1.09	0.27	-0.02	0.07
(8 1) vs (7 1)	0.08	0.02	3.55	0.00	0.04	0.12
(9 0) vs (7 1)	0.16	0.02	7.52	0.00	0.12	0.20
(9 1) vs (7 1)	0.15	0.02	6.54	0.00	0.10	0.19
(10 0) vs (7 1)	0.42	0.02	21.32	0.00	0.38	0.46
(10 1) vs (7 1)	0.25	0.03	9.77	0.00	0.20	0.29
(8 1) vs (8 0)	0.05	0.02	2.41	0.02	0.01	0.10
(9 0) vs (8 0)	0.13	0.02	6.26	0.00	0.09	0.18
(9 1) vs (8 0)	0.12	0.02	5.39	0.00	0.08	0.17
(10 0) vs (8 0)	0.39	0.02	19.73	0.00	0.35	0.43
(10 1) vs (8 0)	0.22	0.03	8.69	0.00	0.17	0.27
(9 0) vs (8 1)	0.08	0.02	3.83	0.00	0.04	0.12
(9 1) vs (8 1)	0.07	0.02	3.10	0.00	0.03	0.11
(10 0) vs (8 1)	0.34	0.02	17.49	0.00	0.30	0.38
(10 1) vs (8 1)	0.17	0.02	6.67	0.00	0.12	0.22
(9 1) vs (9 0)	-0.01	0.02	-0.48	0.63	-0.05	0.03
(10 0) vs (9 0)	0.26	0.02	14.27	0.00	0.22	0.29
(10 1) vs (9 0)	0.09	0.02	3.60	0.00	0.04	0.13
(10 0) vs (9 1)	0.27	0.02	13.48	0.00	0.23	0.31
(10 1) vs (9 1)	0.10	0.03	3.81	0.00	0.05	0.15
(10 1) vs (10 0)	-0.17	0.02	-7.59	0.00	-0.22	-0.13

2 Additional Ecological Predictors at Department Level

2.1 PCGDP

There is a positive and quadratic relationship between PCGDP (Per Capita Gross Domestic Product) and SWB in figure 5, but there is a high spread in SWB (y-axis) along quadratic fit. Among departments with low PCGDP of 2-5k USD, there is a large variability in SWB: ranging from least happy Choco at 7.6 to quite happy Vaupes at 8.3. And then at the high-end of PCGDP of 13-15, there is moderately happy Casanare at 8 to very happy Santander at 8.4. There is also large spread horizontally: many departments with the same SWB on y-axis have very different PCGDP levels on x-axis. Still the relationship is positive and quadratic, as expected and as found in the literature–there are diminishing marginal returns from income in SWB–even in moderately poor Colombia, we find.

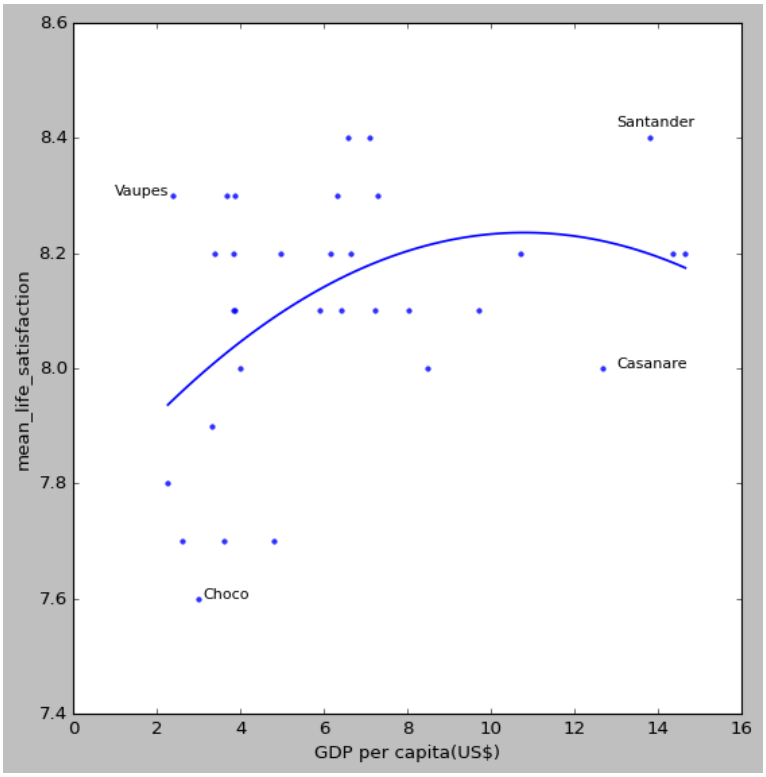


Figure 5

2.2 Population Density

We excluded Bogota because it is not a regular department but capital and San Andres, which is an island. On population density in figure 6 a clear outlier is Atlantico with density at 750,[16] more than double 2nd Quindio at about 300 (not labeled). Most departments are very sparsely populated at below or around 100 people per square kilometer, and there is a great variability in SWB among them from unhappy Choco at 7.6 to several happy departments at 8.4. The denser departments at around 200-300 tend to quite happy at $>$8.1, but some less dense departments are also at $>$8.1. This positive relationship would contradict happy farmer hypothesis, however, departments tend to be large and there is a wide variety of density within, hence the happy farmer hypothesis is not necessarily contradicted.

[16] Atlantico is a relatively small department with large Barranquilla-Soledad metro.

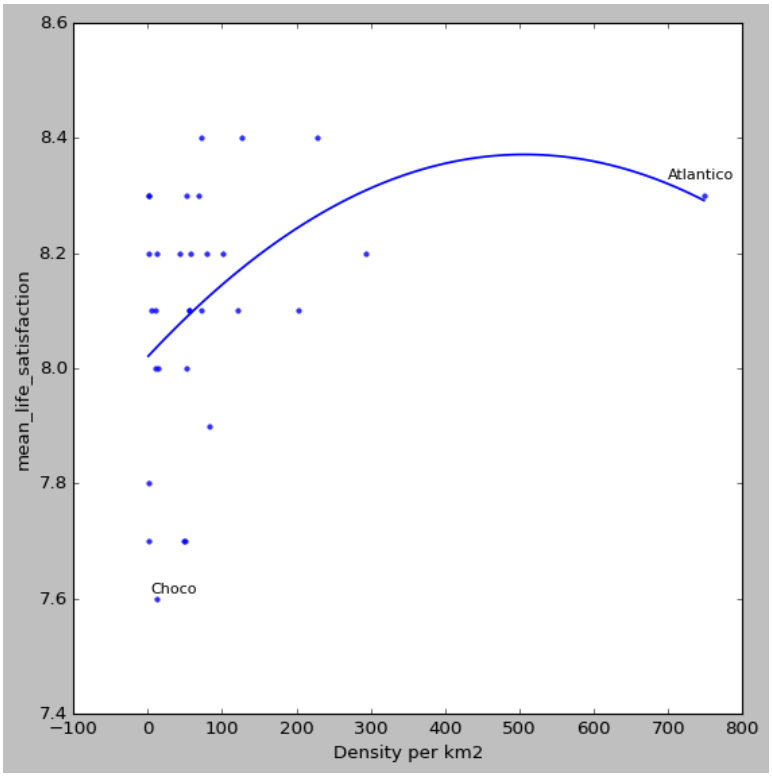


Figure 6

2.3 Crime

The relationship is inverted U-shape in figure 7, opposite to expected negative relationship. However, again, departments tend to be large and there is a wide variety of crime within. And the link between homicide rate in a department and happiness of a person living in that department is much weaker than safety perception of a person and her happiness as studied in the body of the paper.

It is often rather subjective perceptions than objective facts that affect person's wellbeing more.[17]

It may be more of perception of crime as captured by safety satisfaction in the body of the paper than objective crime, which usually disproportionately affects small minority of population such as narcotrafficers and guerrillas. In fact it may be petty crime that upsets an average Colombian more than serious violent crime like homicides. For instance, in Cali while homicide rate has decreased considerably over the years, the petty crime such as petty theft remains high and bothersome for an average Colombian (Martínez and Short 2020).

[17] As the saying goes, "the city on the ground and the city in the mind" (Michael Pacione)–there is a contrast between the objective, physical reality of a city and the subjective, psychological experience of its inhabitants `https://www.sciencedirect.com/science/article/pii/S0169204602002347`.

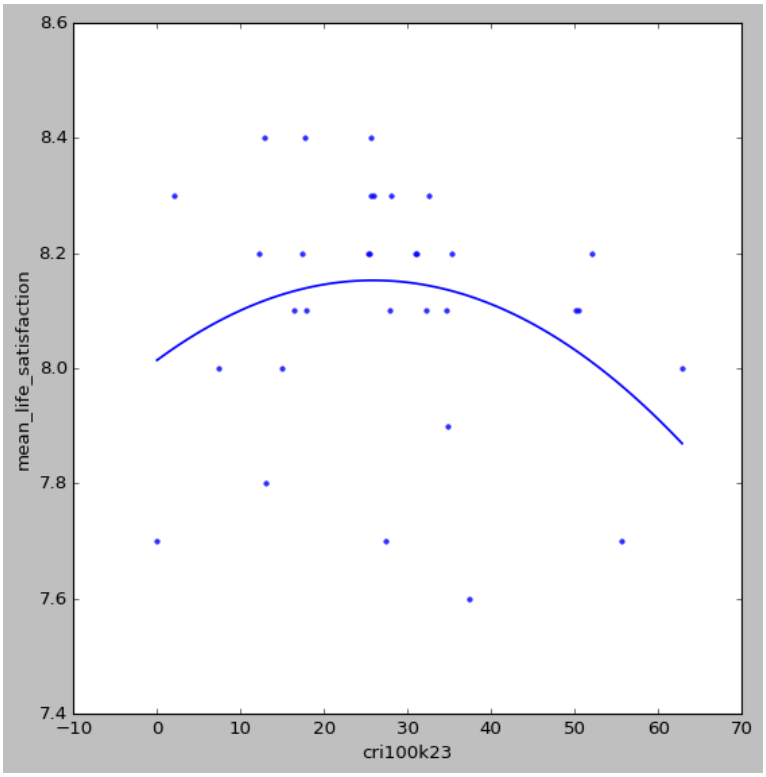


Figure 7

Crime in the West, notably in the US, tends to be urban, but in Colombia in addition to high urban crime, there is also also high rural crime.

Large Colombian cities have its ghettos with rampant crime. And rural geography of crime has to do with infrastructure and climate. Infrastructure: roads used to transport drugs and sea ports used to ship it. At low altitude in warm climate there is coca cultivation, but at high altitudes in cold climate it is more peaceful as one can only farm onions, potatoes, and so forth.

3 Re-estimation Using 2021 ECV

2021 May be biased say because of covid, especially that it peaked in Colombia in 2021 `https://coronavirus.jhu.edu/region/colombia`(by 2023 used in the paper it pretty much cleared). But it a useful robustness check, if covid didn't change the results much, then the results should be pretty stable. Also note that covid in Colombia had quite high mortality `https://coronavirus.jhu.edu/data/mortality`

3.1 Departments/Provinces

Again as said in the paper, overall results are similar to 2023 reported in the body of the paper, except both small sample Amazonas and Vaupés, South-Eastern and with 8.3 in 2023 v 7.6 in 2021, a .7 difference. Second largest differences are for Santander and Putumayo which increased by .4 from 2021 to 2023.

Table 2: Life satisfaction by department.

department	life satisfaction 2023	total obs 2023	non-missing life satisfaction 2023	mean life satisfaction 2021	total obs 2021	non missing life satisfaction 2021	2023-2021
Amazonas	8.3	276	245	7.6	423	365	0.7
Antioquia	8.2	5499	4729	8.3	6036	5060	-0.1
Arauca	8.1	974	773	8.1	1103	868	0
Atlántico	8.3	3144	2311	8.0	2338	1799	0.3
Bogotá, D.C.	8.2	4688	3446	8.1	5337	3810	0.1
Bolívar	8.2	2584	2258	8.1	2495	2156	0.1
Boyacá	8.0	4129	3649	7.8	4638	4122	0.2
Caldas	8.4	2822	2555	8.2	3335	3005	0.2
Caquetá	8.1	2547	2018	7.9	2846	2249	0.2
Casanare	8.0	1749	1370	8.1	2051	1457	-0.1
Cauca	7.7	2580	2244	7.9	2572	2287	-0.2
Cesar	8.3	2224	1717	8.0	1833	1510	0.3
Chocó	7.6	857	754	7.9	1016	913	-0.3
Cundinamarca	8.1	3818	3322	7.8	4233	3684	0.3
Córdoba	8.1	2706	2240	8.0	2226	1933	0.1
Guainía	7.7	327	273	7.4	350	274	0.3
Guaviare	8.2	700	457	8.0	621	411	0.2
Huila	8.1	2961	2322	8.0	2999	2407	0.1
La Guajira	8.2	951	750	7.9	988	736	0.3
Magdalena	8.2	2458	2096	8.0	2110	1710	0.2
Meta	8.2	3683	2752	8.1	3862	2817	0.1
Nariño	7.7	2284	2034	7.4	2847	2492	0.3
Norte de Santander	8.3	2499	2028	8.2	2587	2171	0.1
Putumayo	8.0	950	752	7.6	1272	1012	0.4
Quindío	8.2	2085	1658	8.4	2186	1785	-0.2
Risaralda	8.4	1634	1353	8.3	1996	1663	0.1
San Andrés y Providencia	8.4	28	25	8.2	29	26	0.2
Santander	8.4	3790	3194	8.0	4153	3574	0.4
Sucre	7.9	1798	1399	7.8	1899	1467	0.1
Tolima	8.1	3564	3215	8.0	4257	3770	0.1
Valle del Cauca	8.1	4892	4093	8.1	5268	4421	0
Vaupés	8.3	265	245	7.6	288	267	0.7
Vichada	7.8	515	455	7.6	584	482	0.2

Figure 8 shows 95% CI and hence whether the differences between departments are significant.

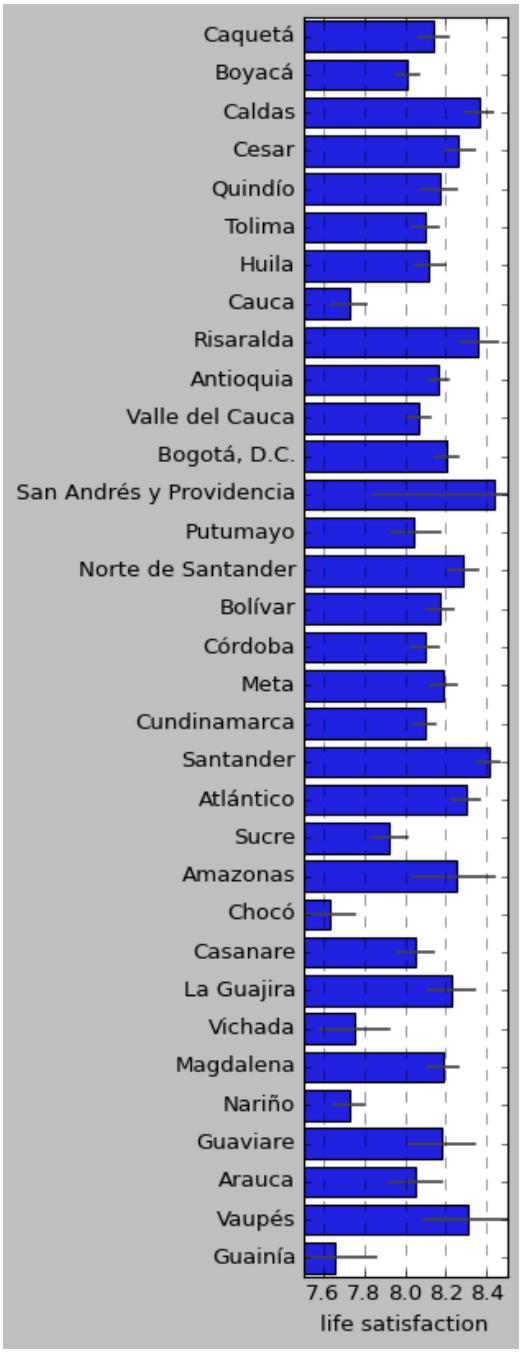


Figure 8: 95% CI shown.

3.2 Cities/Municipalities

This robustness check is very important because ECV is not representative of municipalities.

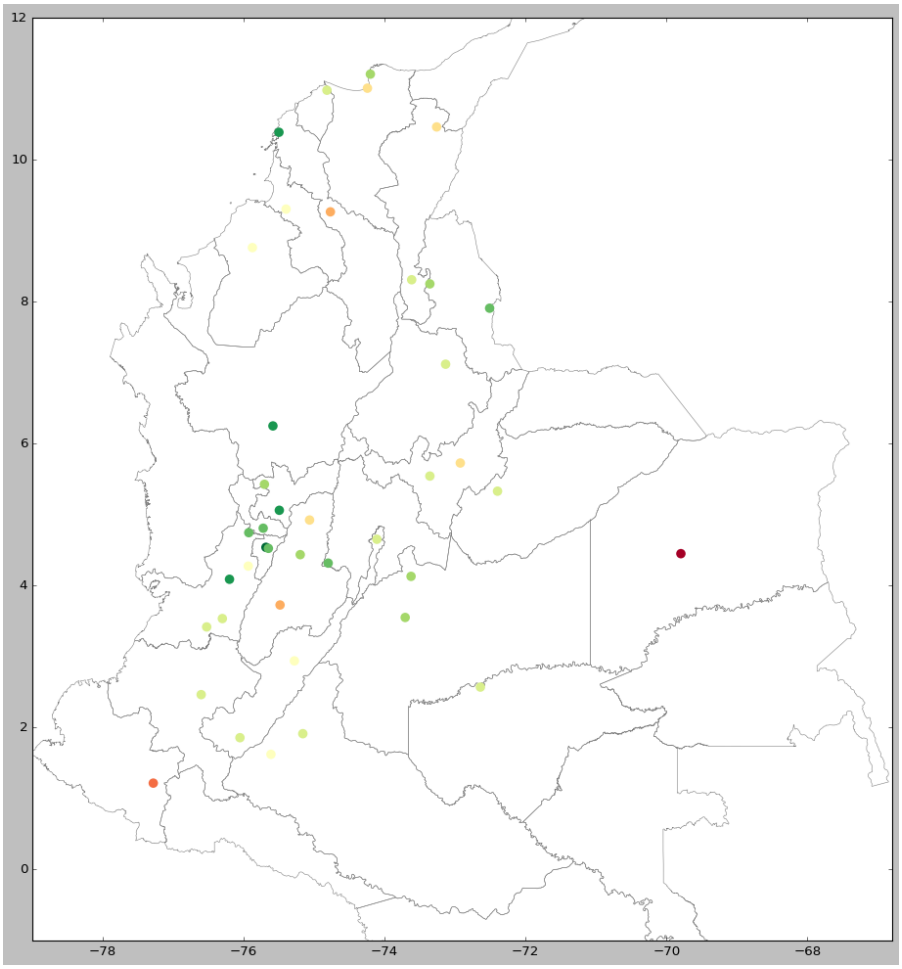


Figure 9: Zoom in to see the detail

Table 3: Life satisfaction by municipality, where non-missing on life satisfaction >200.

municipality name	mean life satisfaction 2021	non missing life satisfaction 2021	mean life satisfaction 2023	non missing life satisfaction 2023	2023-2021 diff
MEDELLÍN	8.4	959.0	8.4	905.0	0.0
BARRANQUILLA	8.1	1116.0	8.3	1363.0	0.2
SABANALARGA	NaN	NaN	8.3	234.0	NaN
BOGOTÁ, D.C.	8.1	3836.0	8.2	3446.0	0.1
CARTAGENA DE INDIAS	8.4	572.0	8.2	591.0	-0.2
EL CARMEN DE BOLÍVAR	NaN	NaN	7.8	200.0	NaN
MAGANGUÉ	7.8	207.0	8.3	205.0	0.5
TUNJA	8.1	241.0	7.9	270.0	-0.2
SOGAMOSO	7.9	367.0	8.2	236.0	0.3
MANIZALES	8.4	567.0	8.4	462.0	0.0
RIOSUCIO	8.2	203.0	8.5	203.0	0.3
FLORENCIA	8.0	630.0	8.2	575.0	0.2
POPAYÁN	8.1	232.0	7.9	225.0	-0.2
VALLEDUPAR	7.9	274.0	8.0	339.0	0.1
AGUACHICA	8.1	215.0	8.6	202.0	0.5
MONTERÍA	8.0	372.0	8.3	465.0	0.3
LORICA	NaN	NaN	8.1	201.0	NaN
GIRARDOT	8.3	222.0	8.1	200.0	-0.2
NEIVA	8.0	436.0	8.3	417.0	0.3
PITALITO	8.1	288.0	8.0	293.0	-0.1
SANTA MARTA	8.2	218.0	8.3	225.0	0.1
CIÉNAGA	7.9	252.0	8.4	234.0	0.5
EL BANCO	NaN	NaN	8.3	200.0	NaN
VILLAVICENCIO	8.2	990.0	8.2	982.0	0.0
GRANADA	8.2	233.0	8.3	229.0	0.1
PASTO	7.7	309.0	7.9	244.0	0.2
SAN JOSÉ DE CÚCUTA	8.3	553.0	8.3	573.0	0.0
ARMENIA	8.5	681.0	8.1	668.0	-0.4
CALARCÁ	8.3	257.0	8.4	244.0	0.1
PEREIRA	8.3	677.0	8.4	524.0	0.1
BUCARAMANGA	8.1	842.0	8.6	798.0	0.5
BARRANCABERMEJA	NaN	NaN	8.6	270.0	NaN
SINCELEJO	8.0	275.0	8.1	222.0	0.1
IBAGUÉ	8.2	549.0	8.4	478.0	0.2
SANTIAGO DE CALI	8.1	1108.0	8.0	985.0	-0.1
CARTAGO	8.3	279.0	8.2	220.0	-0.1
PALMIRA	8.1	332.0	7.9	319.0	-0.2
TULUÁ	8.4	307.0	8.2	303.0	-0.2
TAME	NaN	NaN	8.0	203.0	NaN
YOPAL	8.1	240.0	8.3	264.0	0.2
SAN JOSÉ DEL GUAVIARE	8.1	231.0	8.1	246.0	0.0
CUMARIBO	7.5	215.0	7.5	225.0	0.0

Figure 10 shows 95% CI and hence whether the differences between departments are significant.

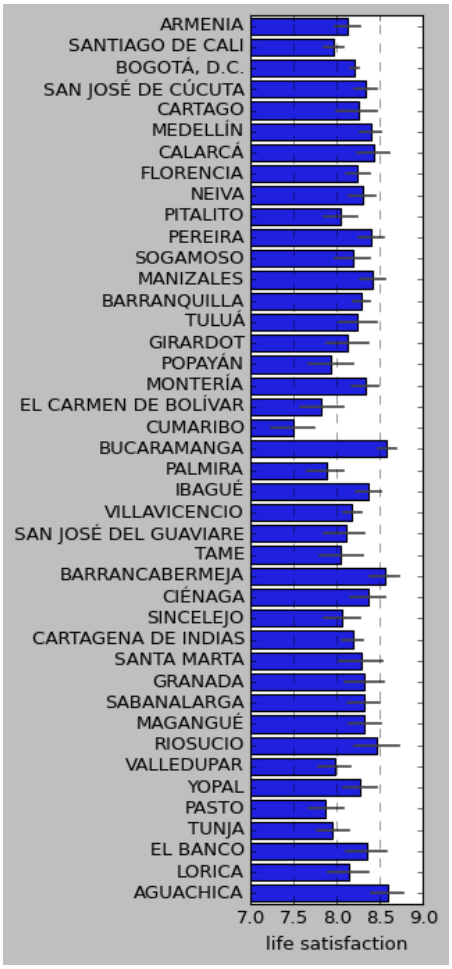


Figure 10: 95% CI shown.

3.3 Urban-Rural and Farmers

Colombia is quite urbanized at around 70-80%. `https://population.un.org/wup/countryprofiles?country=Colombia&mainCategory`
`Countries%20and%20Aggregates&subCategory=Urbanization%20-%20International%20definition`

Still, a substantial part at 20-30% is rural.

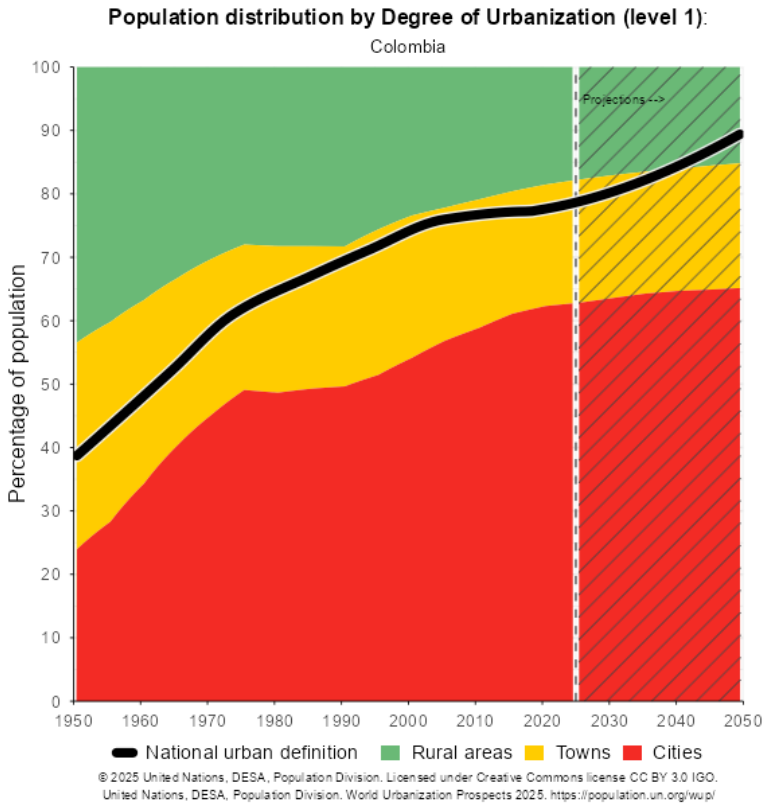


Figure 11

For rural in 2021 pattern is similar to 2023 in body of the paper but less strong for lower deciles where rural are happier than urban, but stronger for 2 top deciles where the opposite is true. For farmers in 2021 there is no happiness premium in lower deciles, but for non-farmers there is happiness premium in 2 top deciles.

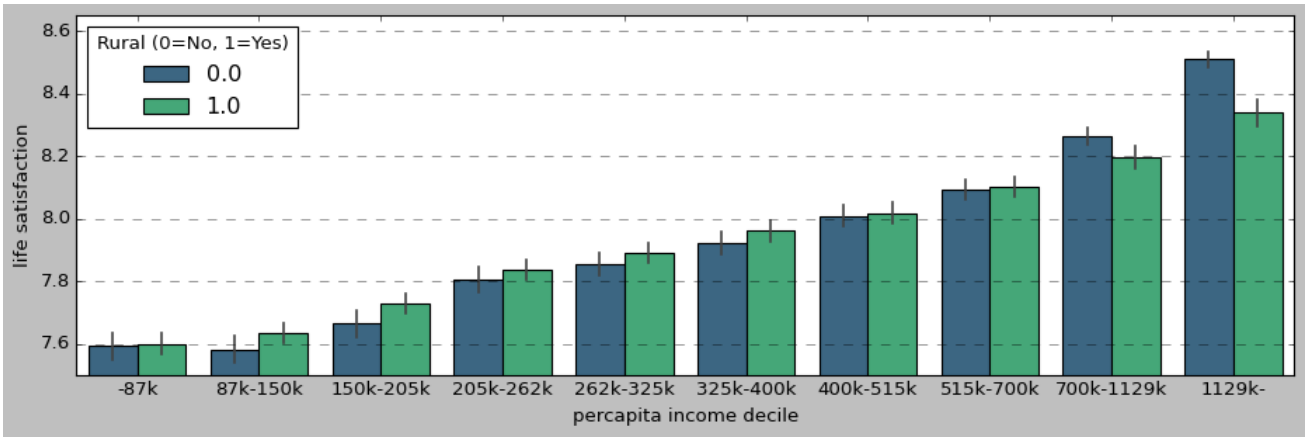


Figure 12

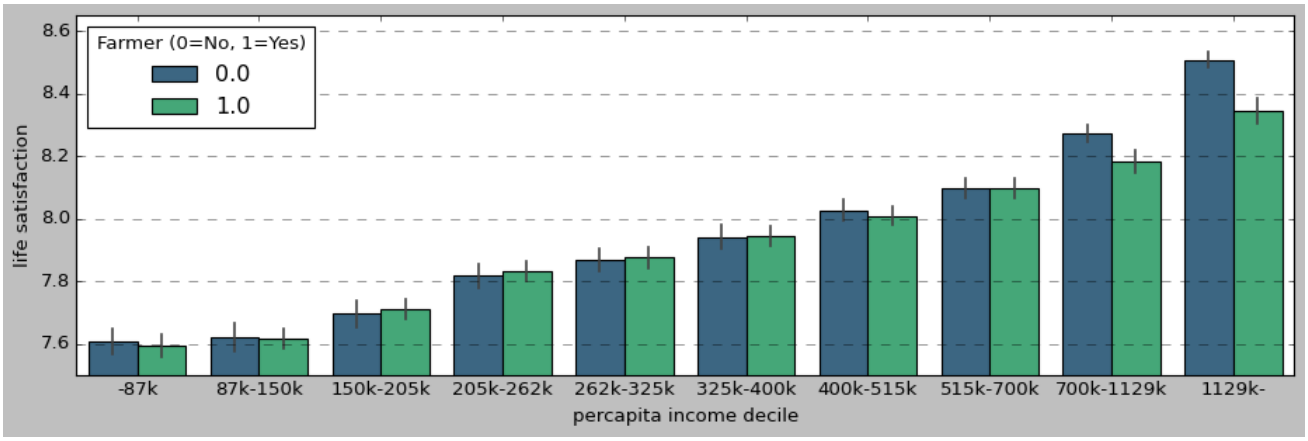


Figure 13

4 Geography of Colombia

4.1 Administrative Boundaries

Colombia is at North-West of South America (fig 14), a large country of about 54m, third largest in Latin America after Brazil and Mexico (`https://fred.stlouisfed.org/series/POPTOTCOA647NWDB`.) Colombia is also large in land mass at about 1.1m sq km, while much smaller than Brazil at 8.5m sq km, it is larger than Venezuela at .9m sq km, and about 3x larger than Germany at .35m sq km and about 2x of Spain at .5 sq km.



Figure 14: Colombia circled in red in Latin America shaded in green.

Map of the 32 departments is in figure 15.



Figure 15: Colombia 32 departments. Source: `https://upload.wikimedia.org/wikipedia/commons/thumb/6/67/Departments_of_colombia.svg/960px-Departments_of_colombia.svg.png`.

4.2 Physical Geography

The paper is about happiness geography, but it would be incomplete without physical geography. Topography is well mapped and best viewed at `https://en.wikipedia.org/wiki/File:Mapa_de_Colombia_(relieve).svg`, where you can zoom in to see the detail. Population is nicely visualized in a 3d map at `https://geoportal.dane.gov.co/geovisores/sociedad/grilla-dane-vihope/`. To compare topography v population density here side by side see figure 16. Much of the population is concentrated around mountainous areas (with a notable exception of quite populated and largely flat Caribbean region in North).

Colombia's road infrastructure and connectivity is very poor.[18] Then this could be one explanation for large distinctiveness/uniqueness of regions–they have not become homogeneous as in many other countries as they are weakly connected due to mountains and poor road infrastructure.

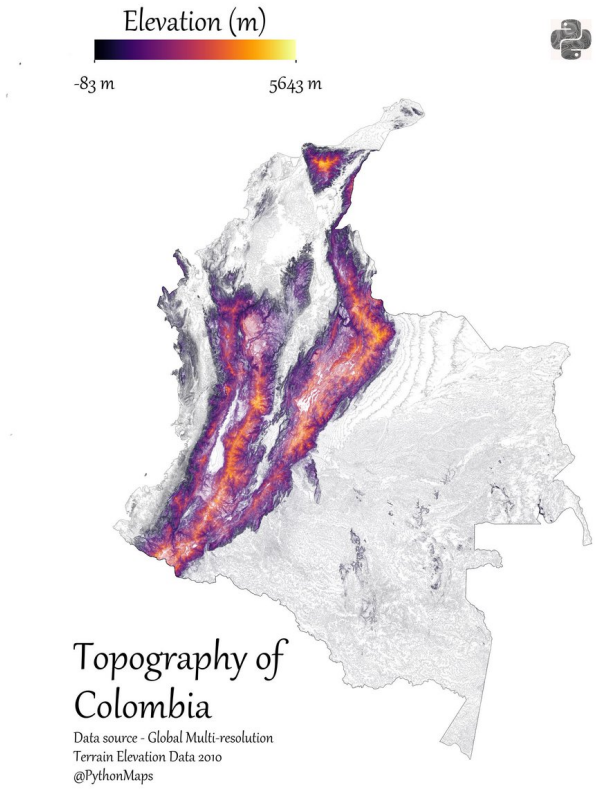
Virtually all of Eastern Colombia, about half of the country, is uninhabited road and city free pristine nature stretching from Eastern Arauca and Vichada down to Amazonas. Likewise, but much narrower, pristine nature belt stretches over much of Western Pacific, especially in Choco.

Colombian population is concentrated along 2 valleys in Andes–one along Panamericana highway from second largest Medellin through Cafetero region, third largest Cali, down to Ipiales on border with Ecuador. The second valley runs from Tunja to largest Bogota down to Neiva. And there is also quite populated Caribbean.

[18] 2018 quality of roads rank is 110/137 (`https://reports.weforum.org/pdf/gci-2017-2018-scorecard/WEF_GCI_2017_2018_Scorecard_EOSQ057.pdf`)–part of the problem is mountains.

Figure 16: Topography and population density. Both maps come from visually appealing PythonMaps: hi-res versions are at the below links. A 3d topography map is also at `https://www.reddit.com/r/MapPorn/comments/u9gr9j/the_topography_of_colombia`. And another topography map at `https://upload.wikimedia.org/wikipedia/commons/3/3b/Colombia_Topography.png`.

(a) Topography `https://x.com/PythonMaps/status/1663937378782937088/photo/1`



(b) Population density `https://x.com/PythonMaps/status/1699089505133772803/photo/1`



Last but not least ecological footprint. `https://data.footprintnetwork.org/` is a useful global overview, Colombian footprint per person is quite low at 2gha, 4x lower than that of the US at 8 gha, and it is stable since 1960s, but bio-capacity per person has declined from 11.4 gha in 1960 to 3.3 in 2024. `https://happyplanetindex.org/countries/COL/` is another tool for cross-country comparisons.

Figure 17 shows useful hi-resolution visualizations of footprint across Colombia for 2015.[19]

[19]OECD provides more recent update across Colombia: `https://www.oecd.org/en/publications/oecd-environmental-performance-reviews-colombia-2026_968398f7-en/full-report/biodiversity-conservation-and-sustainable-use_2099ac2c.html`

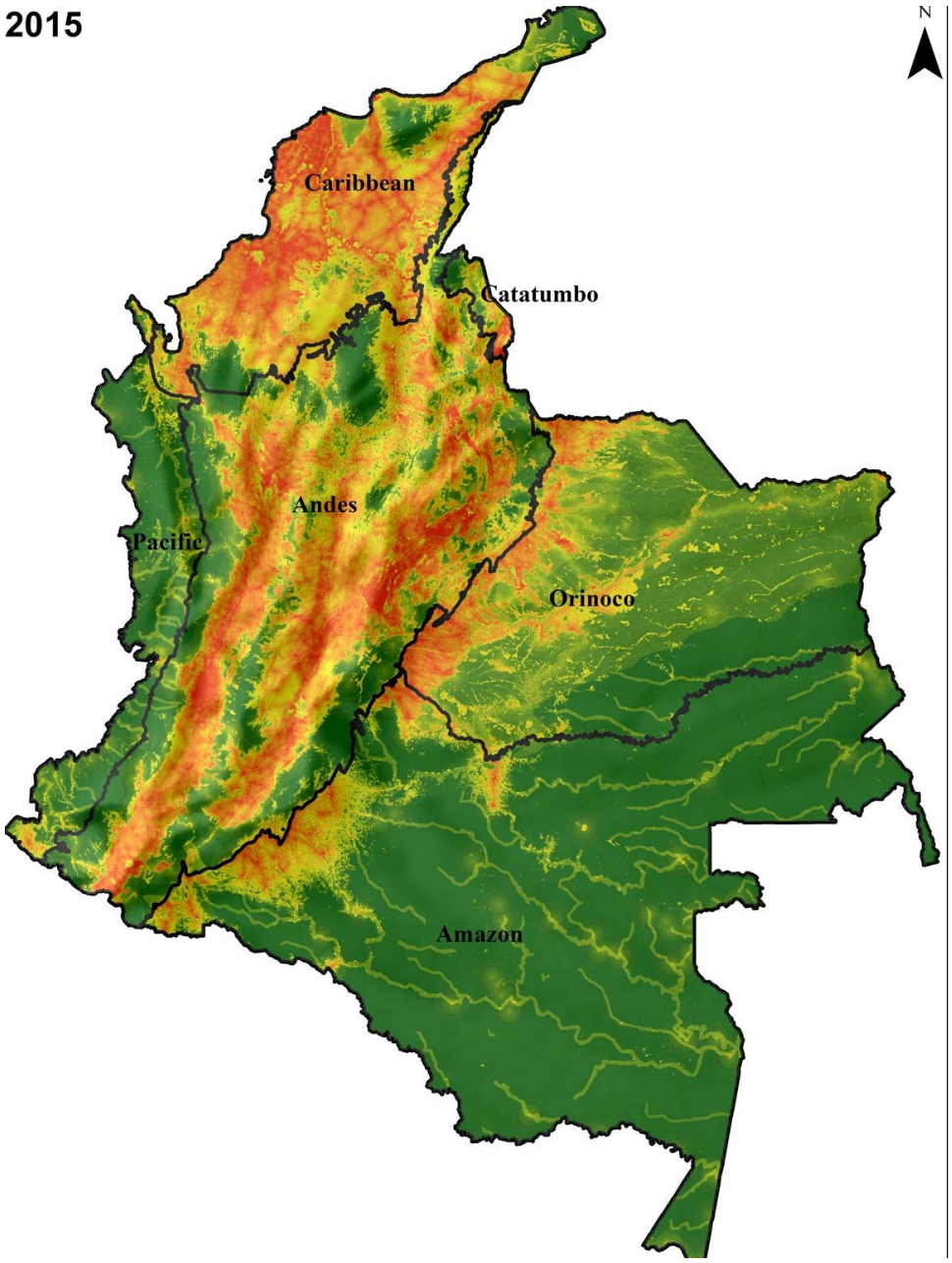
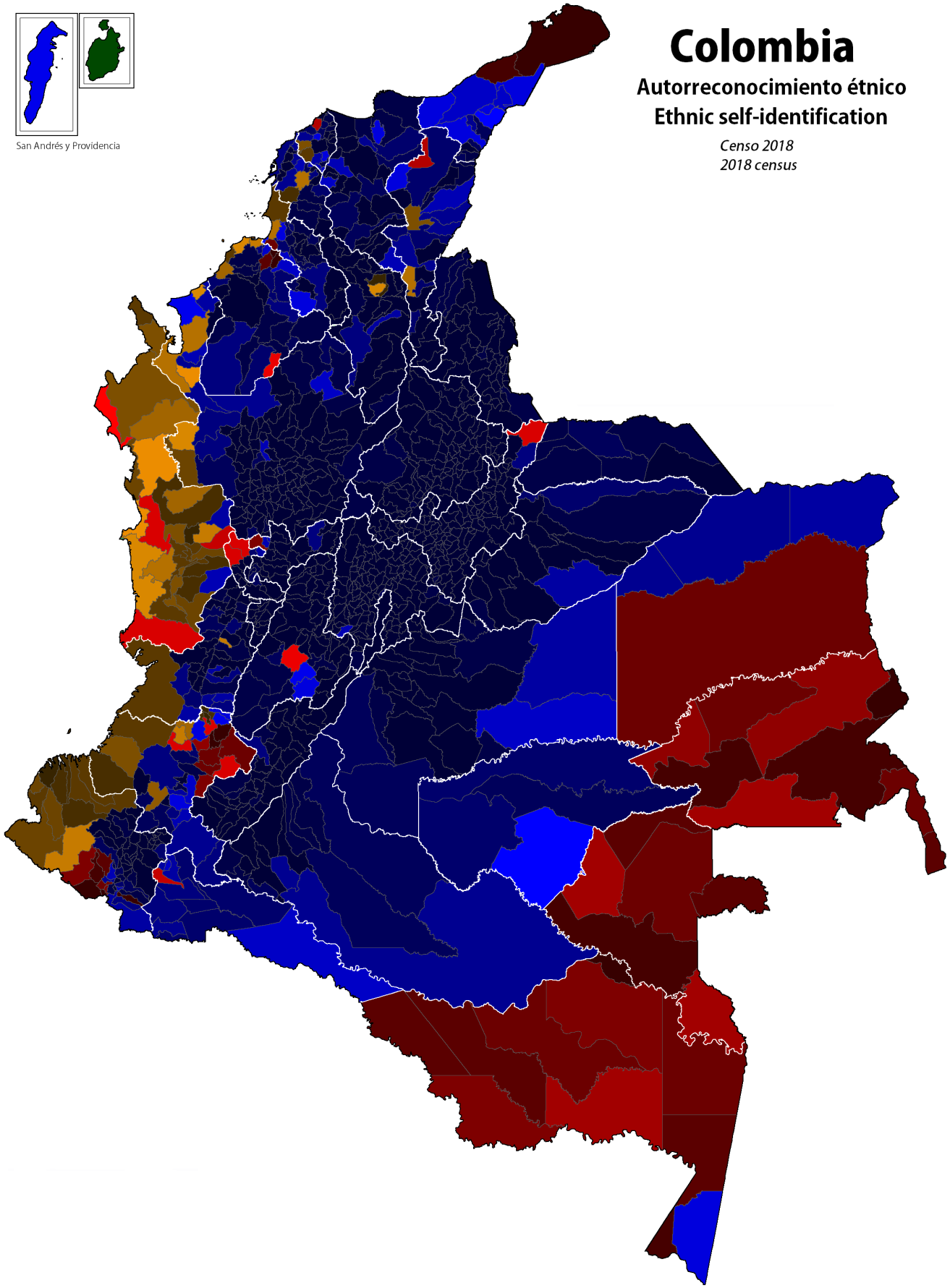


Figure 17: Human footprint: Low-Medium-High: Green-Yellow-Red. Source: `https://www.sciencedirect.com/science/article/pii/S1470160X20305677`

4.3 Socio-cultural Geography

Ethnic self-identification is shown in figure 18.


Colombia
Autorreconocimiento étnico
Ethnic self-identification
Censo 2018
2018 census
San Andrés y Providencia
>30% >35% >40% >45% >50% >55% >60% >65% >70% >75% >80% >85% >90% >95%
Mestizos y blancos
Afrocolombiano
Indígena
Raizal
Gaël Malo L'Hermine
@glhermine
Ottawa, ON (2020)

Colombia has extraordinary diversity, not just biodiversity (2nd largest in the world after 8x larger Brazil), but also cultural and social. Colombia can be thought of as a microcosm of Latin America culturally and socially[20] with Indigenous, African, and Spanish heritage. Indigenous heritage and culture is still alive and widespread, as opposed to the US where it was largely eliminated (Whittle 2026).

Administratively Colombia has 32 departments further divided into municipalities. They are very diverse, almost as if different countries. And there are large differences within departments and across their municipalities as well. For instance in Narino, Pasto is very different from Tumaco. Likewise, in Valle de Cauca, Cali is very different from Buenaventura.

Urbanization differs widely–and urbanization is not just built environment by way of life (Wirth 1938). Again about half of eastern Colombia is wilderness, then there are many small towns and villages with slow and communal way of life, and then there are also a handful of cities where way of life is more Western-like.

For instance, Bogota (or at least parts of it) is similar to other large fast paced commerce oriented cities like New York or London. Likewise, it is very international and has the most connected airport in Latin America (most international connections) and one of the 3 largest airports in Latin America (close to Sao Paulo and Mexico city).[21] Cartagena is Miami-like: expensive and fashionable catering to the rich and tourists. On the other hand, Cali, the third largest city, is very much Latin with more laid back atmosphere and salsa music. Indeed, one could think these cities belong to different countries or even cultures. Pacific Buenaventura and Tumaco are different yet–less organized and impoverished. One could enumerate many more examples of differences across municipalities.

Bogota could be considered a primate city, although the difference versus second largest Medellin is only of order of about 2, smaller than in many other countries[22]

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[20] See https://colombia.co/en/colombia-country/history/greatest-cultural-blend and Okulicz-Kozaryn and Martinez (2025). Also physically–the only South American country with both Pacific and Atlantic coasts, and has just about any type of topography, climate, and vegetation: Andes with several snowy peaks over 5km, hot and dry deserts, humid rain forests, and moderate climate valleys.

[21] https://www.oag.com/blog/latin-americas-most-connected-airports-aviation-infographics and https://en.wikipedia.org/wiki/List_of_the_busiest_airports_in_Latin_America.

[22] https://en.wikipedia.org/wiki/Primate_city.

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